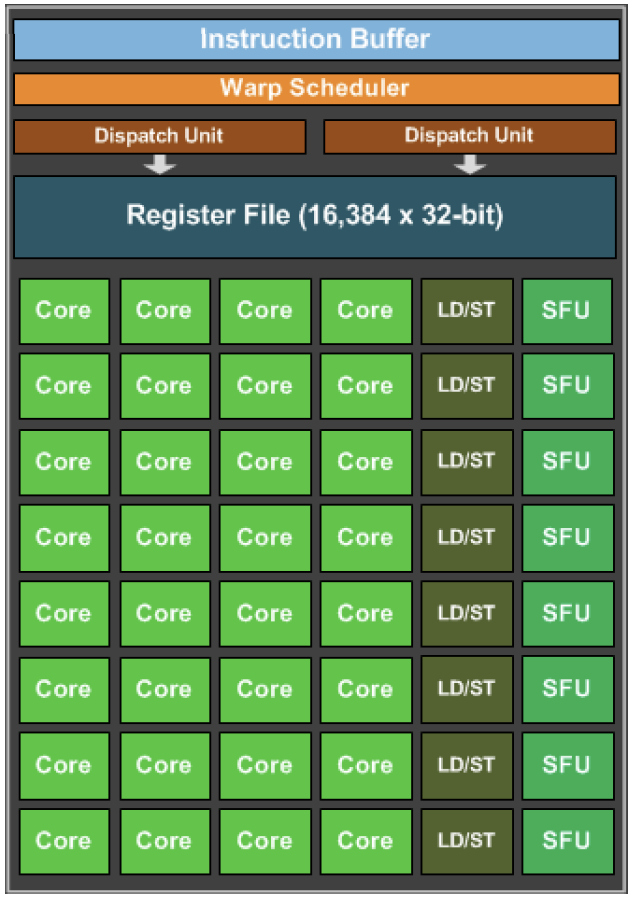
CSE113: Parallel Programming

- **Topics**:
  - Finish up GPUs
  - Homework 4
  - Start on memory models



Announcements

Grading

- Grades for HW 3 should be out by the end of the week

- HW 4 is released.

Kernel constraints

- What can and can’t you do in a GPU kernel?
  - Print?
  - File I/O?
  - C++ standard library? E.g. vectors?
  - Memory allocation?
  - Atomics?

Kernel constraints

- What can and can’t you do in a GPU kernel?
  - Print?
    CUDA now supports printfs directly in the kernel.
  - File I/O?
    You won't be able to implement file I/O directly in CUDA C++, as the GPU does not connect directly to the file system.
  - C++ standard library? E.g. vectors?
    You can't use the STL in CUDA, but you may be able to use the Thrust library to do what you want.
  - Memory allocation?
    In general allocating memory dynamically from the gpu is not possible in CUDA or other GPU APIs.
  - Atomics?
    Every Nvidia GPU that is a core 84 or higher supports CUDA 1.1 or higher and thus supports atomic operations.

The GPU Program

- If the same GPU also drives the display:
  The operating system expects the GPU kernel to be fast to render graphics.
  It kills the kernel if it takes more than ~5 seconds.

- A headless or second GPU has no limit.

The GPU Program

```
__global__ void vector_add(int * a, int * b, int * c, int size) {
    for (int i = 0; i < size; i++) {
      a[i] = b[i] + c[i];
   }
}
```

calling the function

```
vector_add<<<1,1>>>(d_a, d_b, d_c, size);
```

The GPU Program



1000x slower on GPU?
It didn’t do so well...

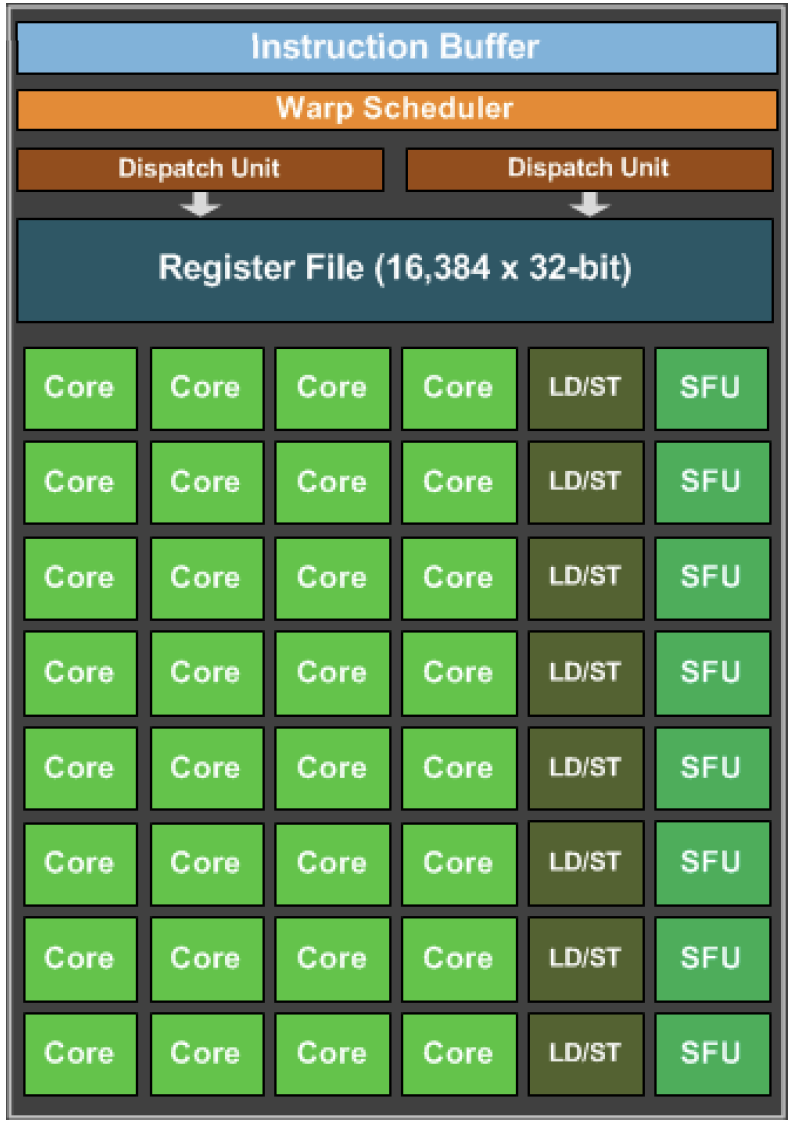
One GPU core is much slower than a single CPU core.

CPU spending silicon on caches, branch prediction and out-of-order logic,
GPU spending it on ALUs and a huge register file.

This saves energy and helps manage many cores.

First parallelization attempt

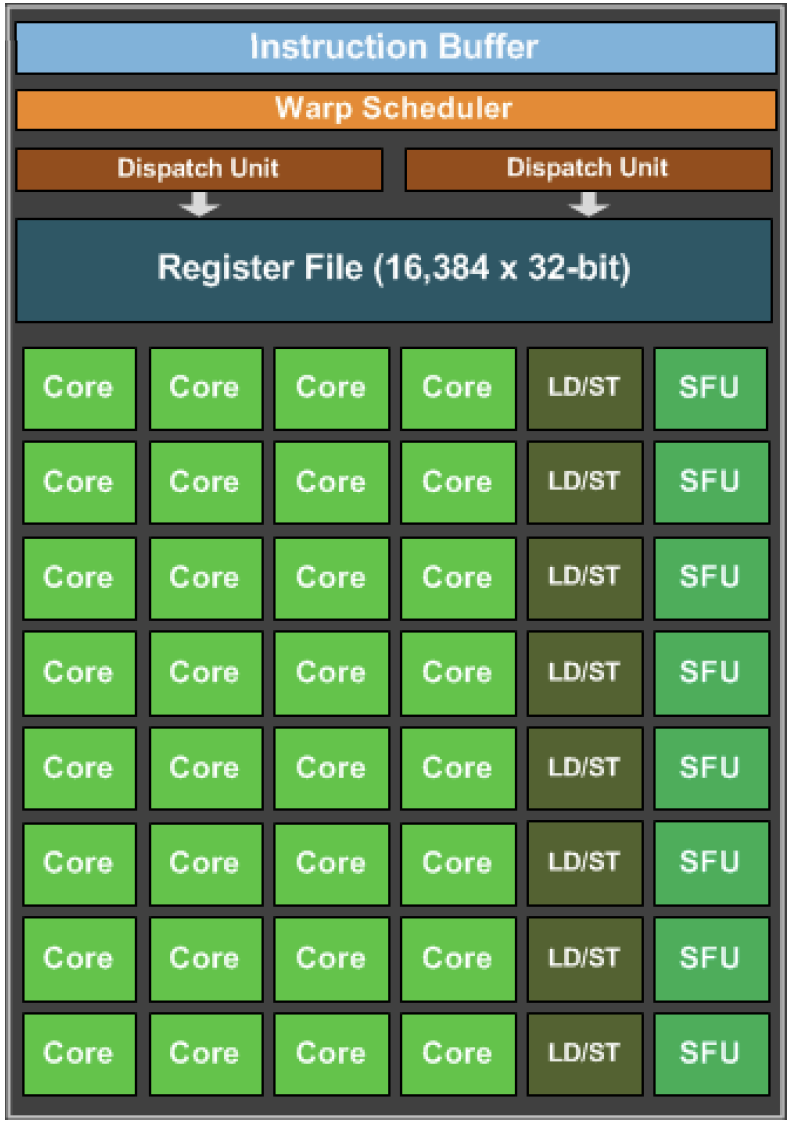
- Let’s look at some GPU documentation.
- The Maxwell whitepaper (Nvidia GPU microarchitecture, the 2014 generation) shows a diagram of one of the GPU cores



https://www.techpowerup.com/gpu-specs/docs/nvidia-gtx-980.pdf

woah, 32 cores!

We should parallelize our application!



https://www.techpowerup.com/gpu-specs/docs/nvidia-gtx-980.pdf

First parallelization attempt

```
__global__ void vector_add(int * d_a, int * d_b, int * d_c, int size) {
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   }
}
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calling the function

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First parallelization attempt

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    for (int i = 0; i < size; i++) {
      d_a[i] = d_b[i] + d_c[i];
   }
}
```

calling the function

```
vector_add<<<1,32>>>(d_a, d_b, d_c, size);
```

number of threads to launch the program with

First parallelization attempt

```
__global__ void vector_add(int * d_a, int * d_b, int * d_c, int size) {
    int chunk_size = size/blockDim.x;
    int start = chunk_size * threadIdx.x;
    int end = start + chunk_size;
    for (int i = start; i < end; i++) {
      d_a[i] = d_b[i] + d_c[i];
   }
}
```

number of threads

calling the function

```
vector_add<<<1,32>>>(d_a, d_b, d_c, size);
```

First parallelization attempt

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    int end = start + chunk_size;
    for (int i = start; i < end; i++) {
      d_a[i] = d_b[i] + d_c[i];
   }
}
```

number of threads
thread id

calling the function

```
vector_add<<<1,32>>>(d_a, d_b, d_c, size);
```

blockIdx.x is the block id.
blockDim.x is the number of threads per block.
threadIdx.x is the thread id within block.

First parallelization attempt

```
__global__ void vector_add(int * d_a, int * d_b, int * d_c, int size) {
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   }
}
```

calling the function

```
vector_add<<<1,32>>>(d_a, d_b, d_c, size);
```

blockIdx.x is the block id.
blockDim.x is the number of threads per block.
threadIdx.x is the thread id within block.

When you write <<<1, 32>>> the integers are converted to dim3(1,1,1) and dim3(32,1,1), so only the x field is non-trivial and threadIdx.x is the only one you read.

First parallelization attempt

That gives us about 14x speedup!

First parallelization attempt



Getting better but we have a long ways to go!

Quiz 11.

Using a few sentences, give examples of workloads that benefit from GPU parallelism.

Free response: a few sentences on the form

Scan to answer

forms.gle/xcAa74cFCvfMcVhT7

chart: docs.google.com/forms/d/e/1FAIpQLSfafsP0Ovn_31uhRJgjO4RqzZL0ir9-cI74cTtjh88RP87VAQ/viewanalytics

Quiz 11.

Using a few sentences, give examples of workloads that benefit from GPU parallelism.

Free response: a few sentences on the form

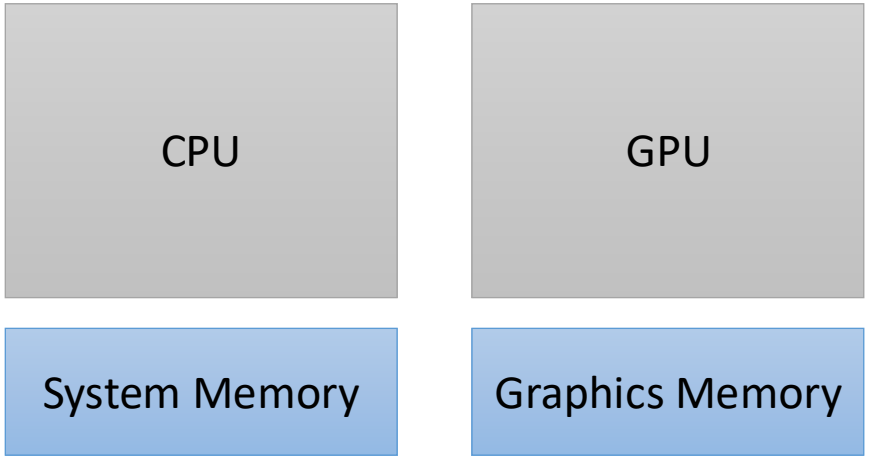
Data parallel.

Scan to answer

forms.gle/xcAa74cFCvfMcVhT7

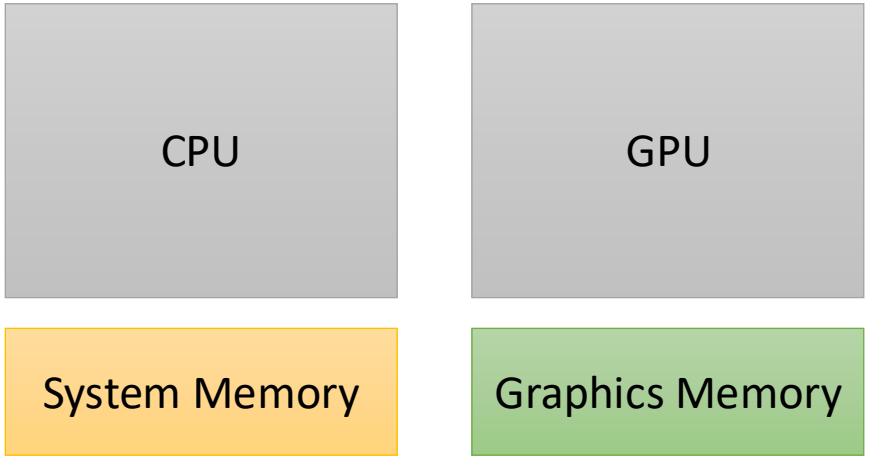
chart: docs.google.com/forms/d/e/1FAIpQLSfafsP0Ovn_31uhRJgjO4RqzZL0ir9-cI74cTtjh88RP87VAQ/viewanalytics

GPU Memory



GPU Memory

CPU Memory:
Fast: Low Latency
Easily saturated: Low Bandwidth
Scales well: up to 1 TB
DDR



GPU Memory:
Slow: High Latency
Hard to saturate: High Bandwidth
Doesn't scale: 80-192 GB
GDDR, HBM

2-lane straight highway driven on by sports cars

Different technologies

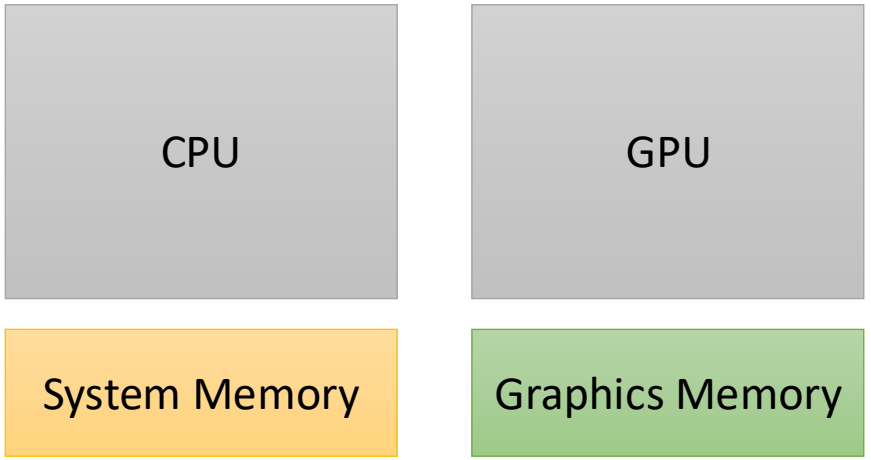
16-lane highway on a windy road driven by semi trucks

GPU Memory

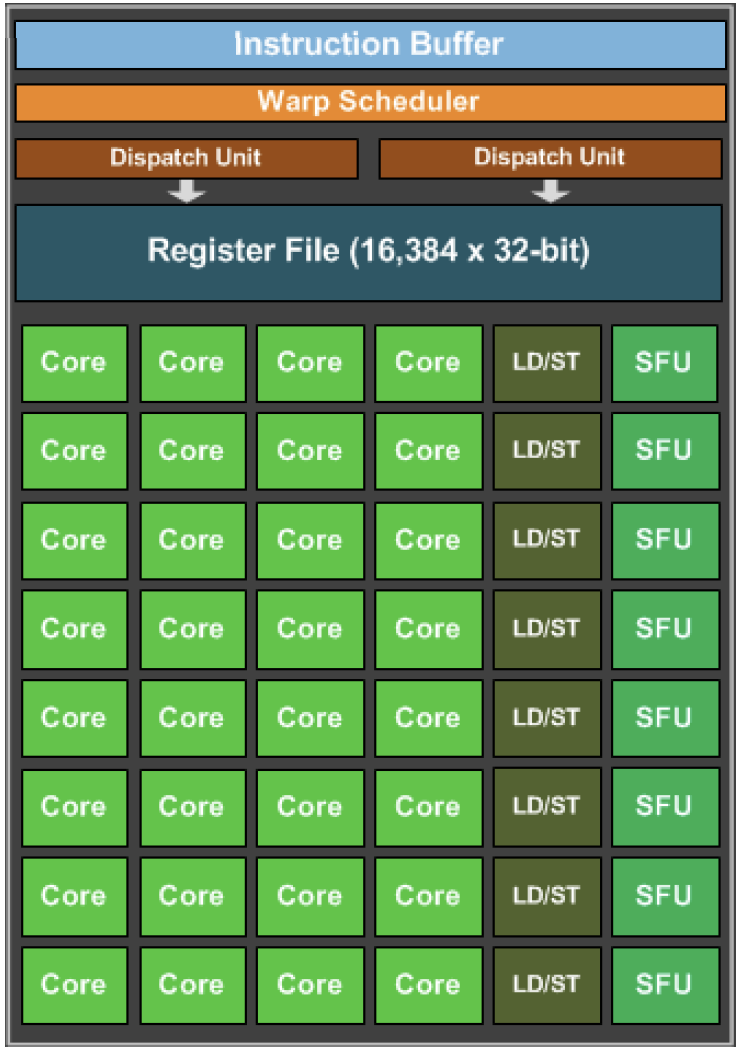
bandwidth:
~**1-3.3 TB/s** for GPU
~**300 GB/s** for CPUs

memory Latency:
~**600** cycles for GPU memory
~**200** cycles for CPU memory

Cache Latency:
~**28** cycles for L1 hit for GPU
~**4** cycles for L1 hit on CPUs



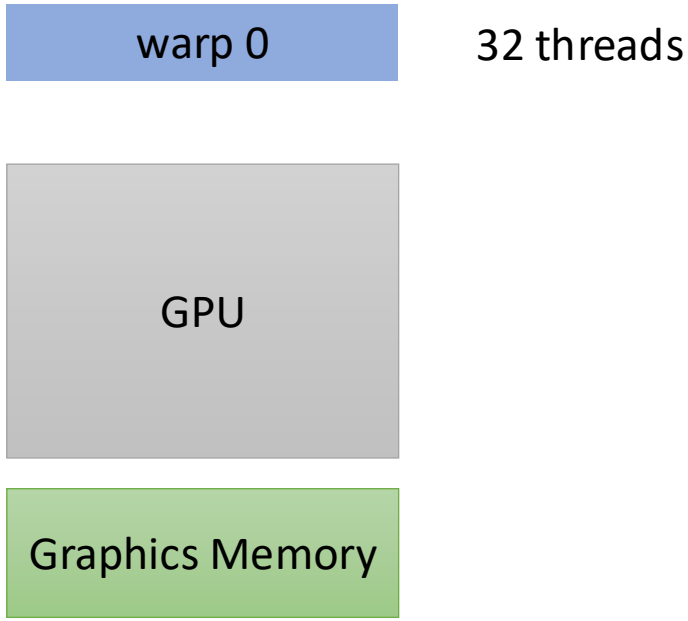
Warp scheduling and concurrency?



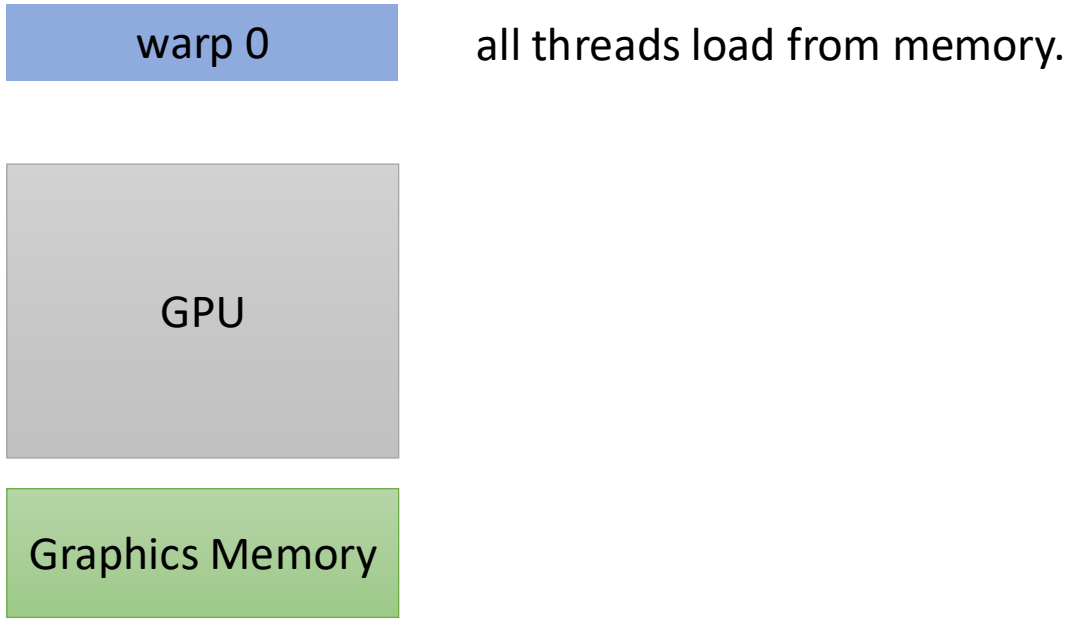
Blocks: multiple warps
Bound on number of warps: 32

Block = 32 warps * 32 blocks = 1024 threads max

Warp scheduling and concurrency?

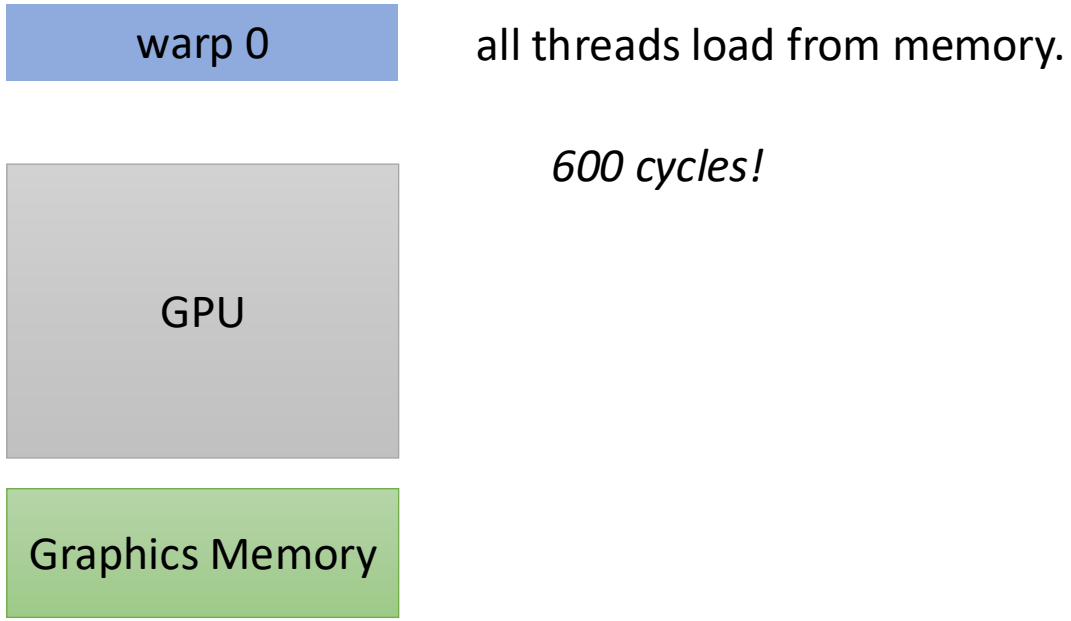


Warp scheduling and concurrency?



all threads load from memory.

Warp scheduling and concurrency?



all threads load from memory.

600 cycles!

Warp scheduling and concurrency?

warp 0

warp 1

warp 2

We can hide latency through scheduling and concurrency!

GPU

Graphics Memory



Warp scheduling and concurrency?

warp 1

warp 2

warp 0

GPU

Graphics Memory



We can hide latency through scheduling and concurrency!

Warp scheduling and concurrency?

memory access
600 cycles

warp 1

warp 2

warp 0

GPU

Graphics Memory



We can hide latency through scheduling and concurrency!

Warp scheduling and concurrency?

memory access
600 cycles

warp 1

warp 2

warp 0

GPU

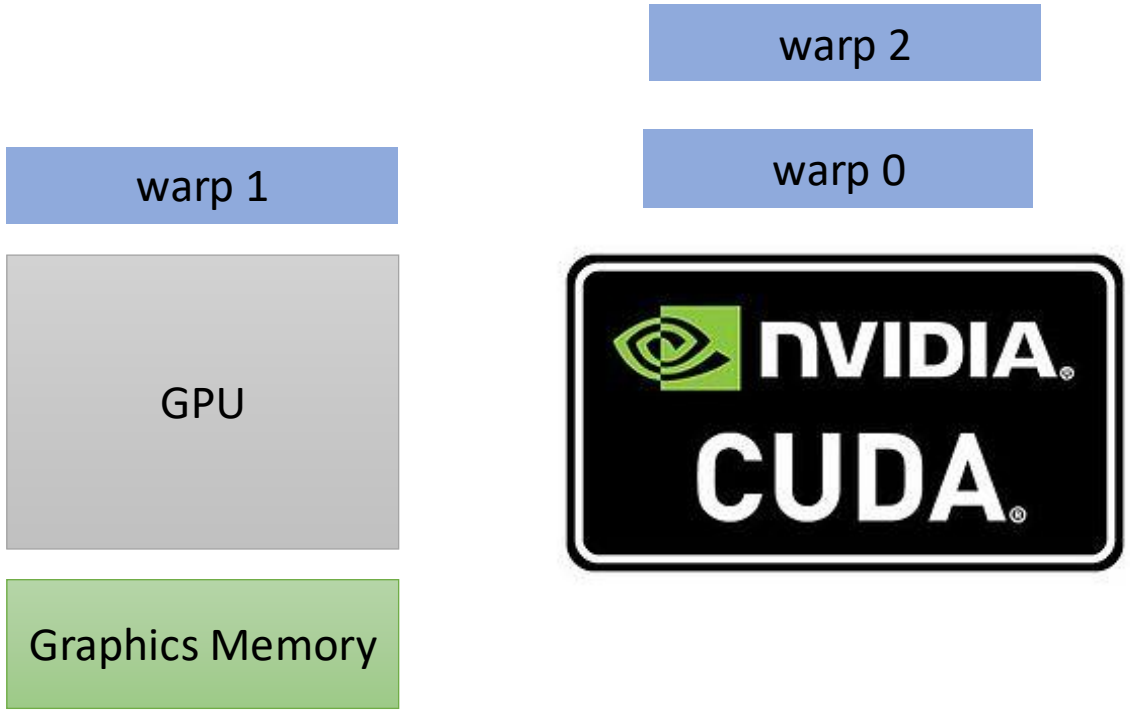
Graphics Memory



preempt warp 0
and put warp 1 on

We can hide latency through scheduling and concurrency!

Warp scheduling and concurrency?



We can hide latency through scheduling and concurrency!

Warp scheduling and concurrency?

memory access
600 cycles

warp 1

GPU

Graphics Memory

warp 2

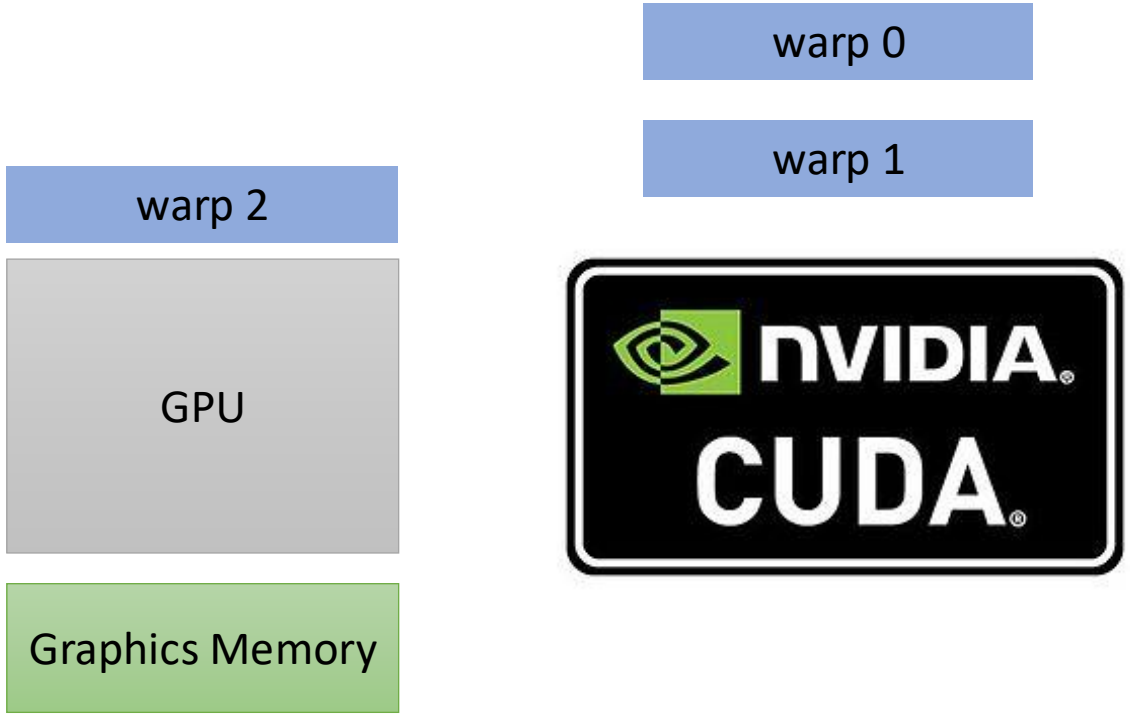
warp 0



preempt warp 1
and put warp 2 on

We can hide latency through scheduling and concurrency!

Warp scheduling and concurrency?



We can hide latency through scheduling and concurrency!

Warp scheduling and concurrency?

memory access
600 cycles

warp 2

GPU

Graphics Memory

warp 0

warp 1



preempt warp 2
and put warp 0 on

We can hide latency through scheduling and concurrency!

Warp scheduling and concurrency?

Hey, my memory has arrived!

warp 0

GPU

Graphics Memory

warp 1

warp 2



preempt warp 2
and put warp 0 on

We can hide latency through scheduling and concurrency!

Quiz 12.

What is a warp?

Select one

A. Group of 32 threads in a GPU

B. Group of 16 threads in a GPU

C. Group of 2 threads in a GPU

D. Group of some threads in a GPU



Scan to answer

forms.gle/vwqLu4z8L7qVbBdd9

chart: docs.google.com/forms/d/e/1FAIpQLSdtiYdVsGaa_pIzyv0wv39CZD-oXhojmrd3oNDMY8qagatO0g/viewanalytics

Quiz 12.

What is a warp?

Select one

A. Group of 32 threads in a GPU

B. Group of 16 threads in a GPU

C. Group of 2 threads in a GPU

D. Group of some threads in a GPU

Nvidia uses 32, AMD uses 64 or 32, Intel varies, and WebGPU calls the same thing a subgroup whose size you query at runtime.

Scan to answer

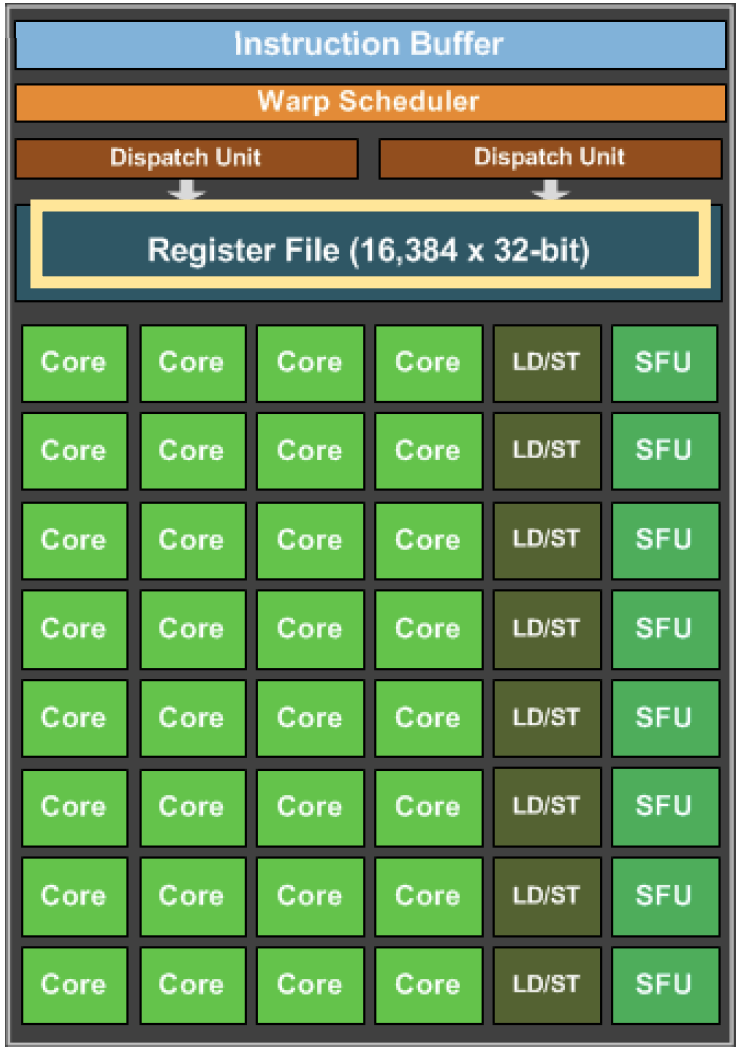
forms.gle/vwqLu4z8L7qVbBdd9

chart: docs.google.com/forms/d/e/1FAIpQLSdtiYdVsGaa_pIzyv0wv39CZD-oXhojmrd3oNDMY8qagatO0g/viewanalytics

Warp scheduling and concurrency?

But wait, I thought rescheduling was expensive?

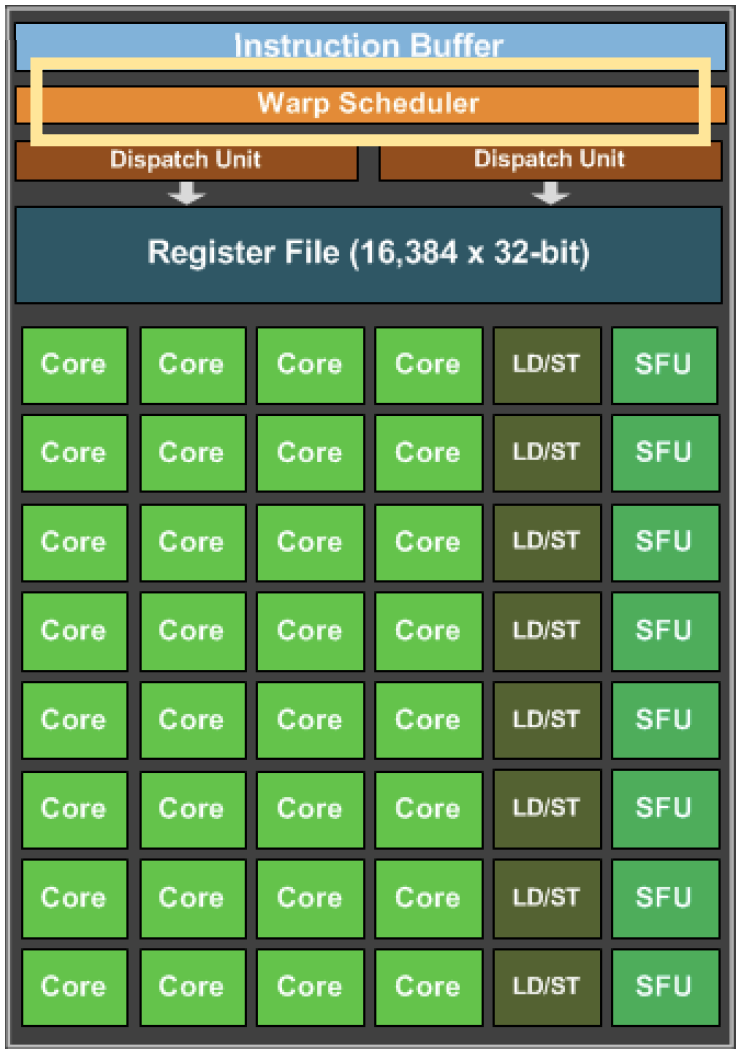
Warp scheduling and concurrency?



But wait, I thought rescheduling was expensive?

Registers all stay on chip. Large register file.

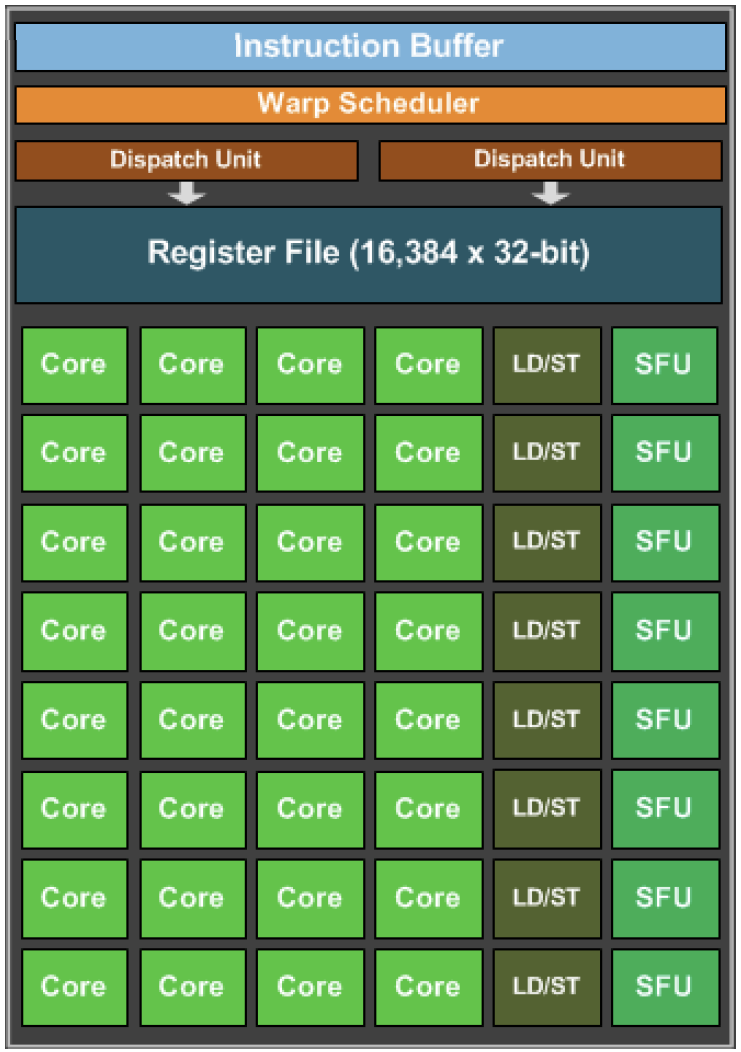
Warp scheduling and concurrency?



But wait, I thought rescheduling was expensive?

dedicated scheduler logic

Warp scheduling and concurrency?



But wait, I thought rescheduling was expensive?
Hardware multithreading, not OS-style preemption.

Go back to our program

```
__global__ void vector_add(int * d_a, int * d_b, int * d_c, int size) {
    int chunk_size = size/blockDim.x;
    int start = chunk_size * threadIdx.x;
    int end = start + chunk_size;
    for (int i = start; i < end; i++) {
      d_a[i] = d_b[i] + d_c[i];
   }
}
```

calling the function

```
vector_add<<<1,32>>>(d_a, d_b, d_c, size);
```

Let's launch with 32 warps
32*32=1024

Go back to our program

```
__global__ void vector_add(int * d_a, int * d_b, int * d_c, int size) {
    int chunk_size = size/blockDim.x;
    int start = chunk_size * threadIdx.x;
    int end = start + chunk_size;
    for (int i = start; i < end; i++) {
      d_a[i] = d_b[i] + d_c[i];
   }
}
```

calling the function

```
vector_add<<<1,1024>>>(d_a, d_b, d_c, size);
```

Let's launch with 32 warps
32*32=1024

Concurrent warps

This is about 2x faster.

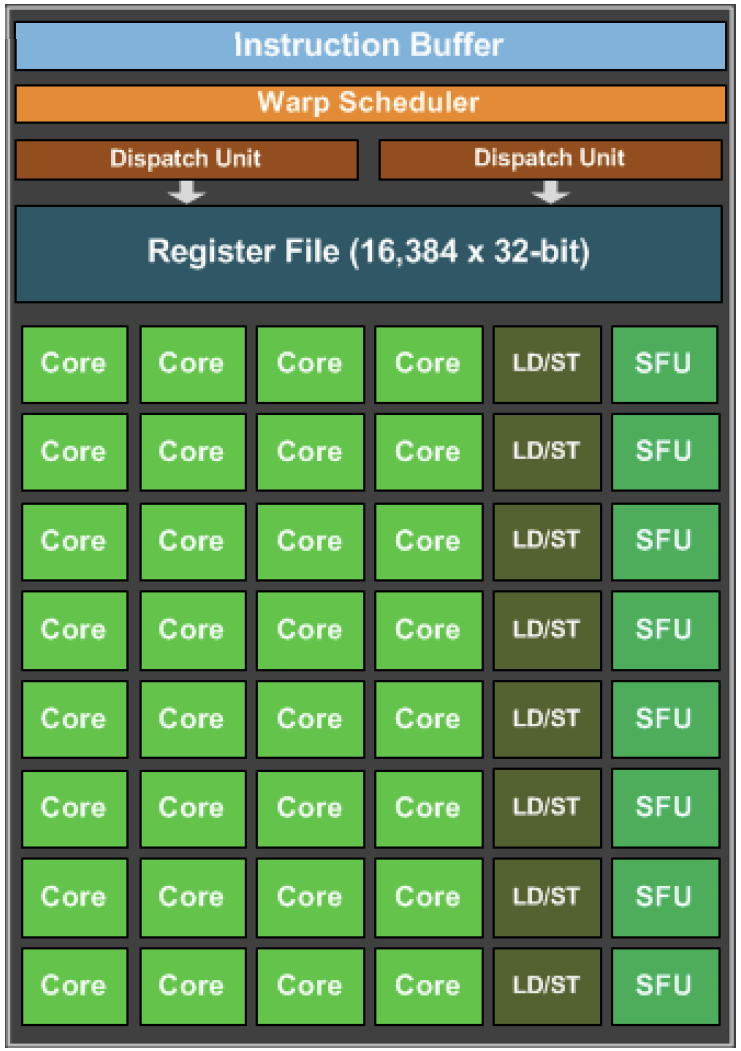
Concurrent warps

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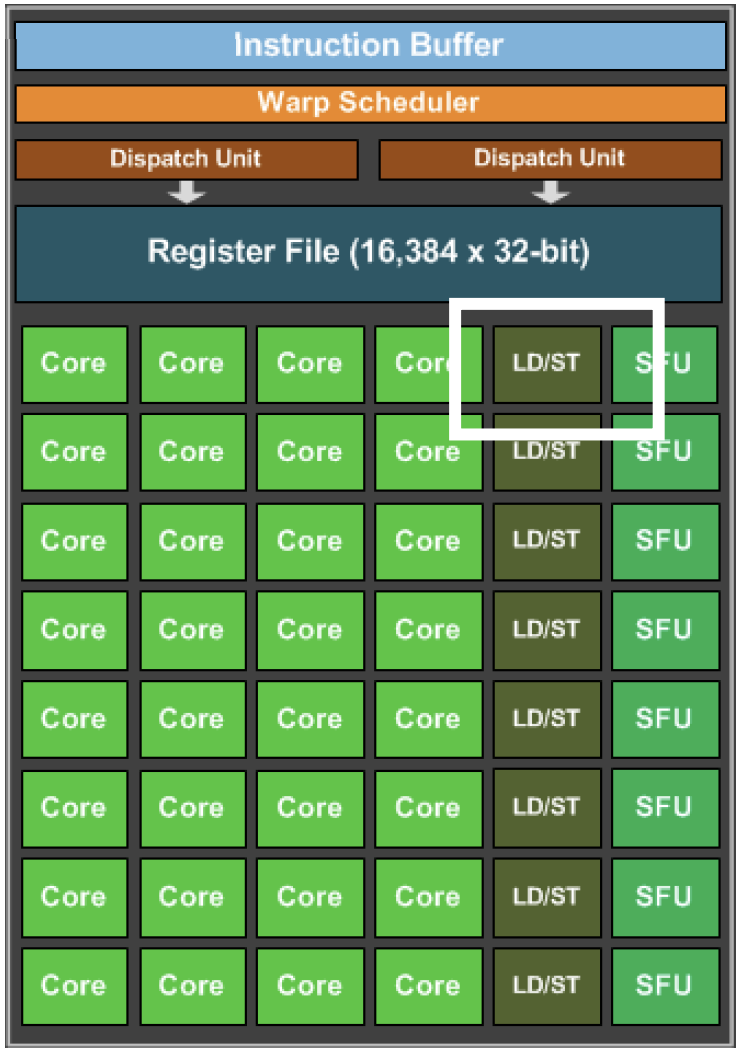


Getting better!

Optimizing memory accesses

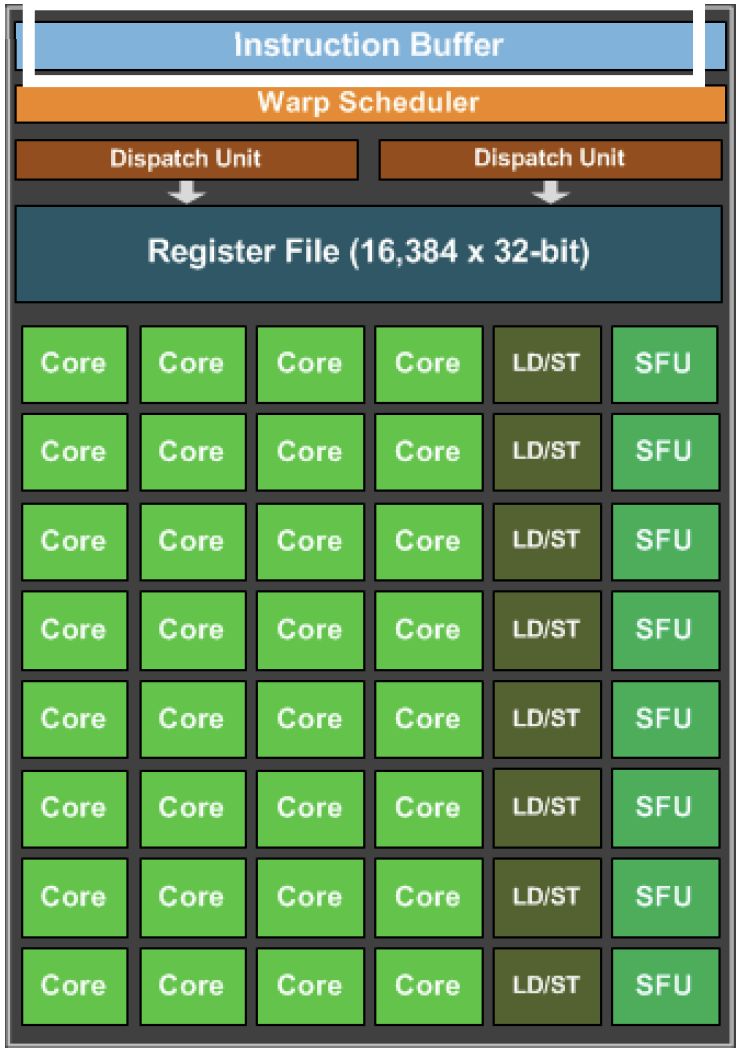


Optimizing memory accesses



This is the load/store unit. The hardware component responsible for issuing loads and stores.
Shared between 4 cores.

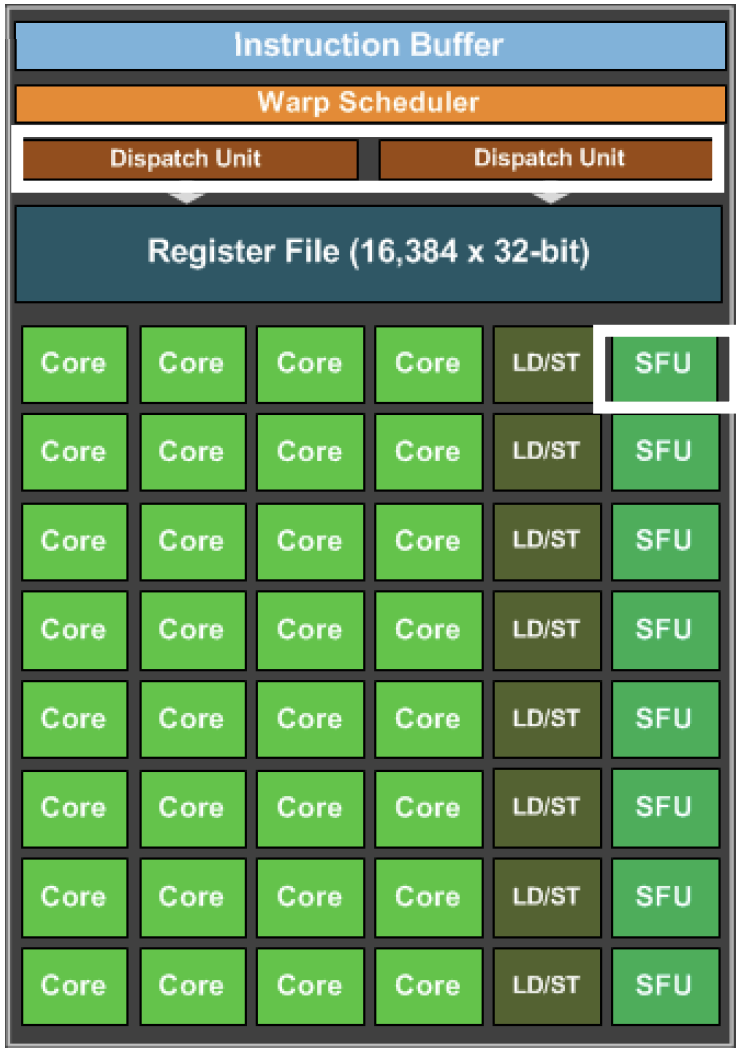
Optimizing memory accesses



This is the instruction cache
Shared between all cores of the warp.

This is the load/store unit. The hardware component responsible for issuing loads and stores.
Shared between 4 cores.

Optimizing memory accesses

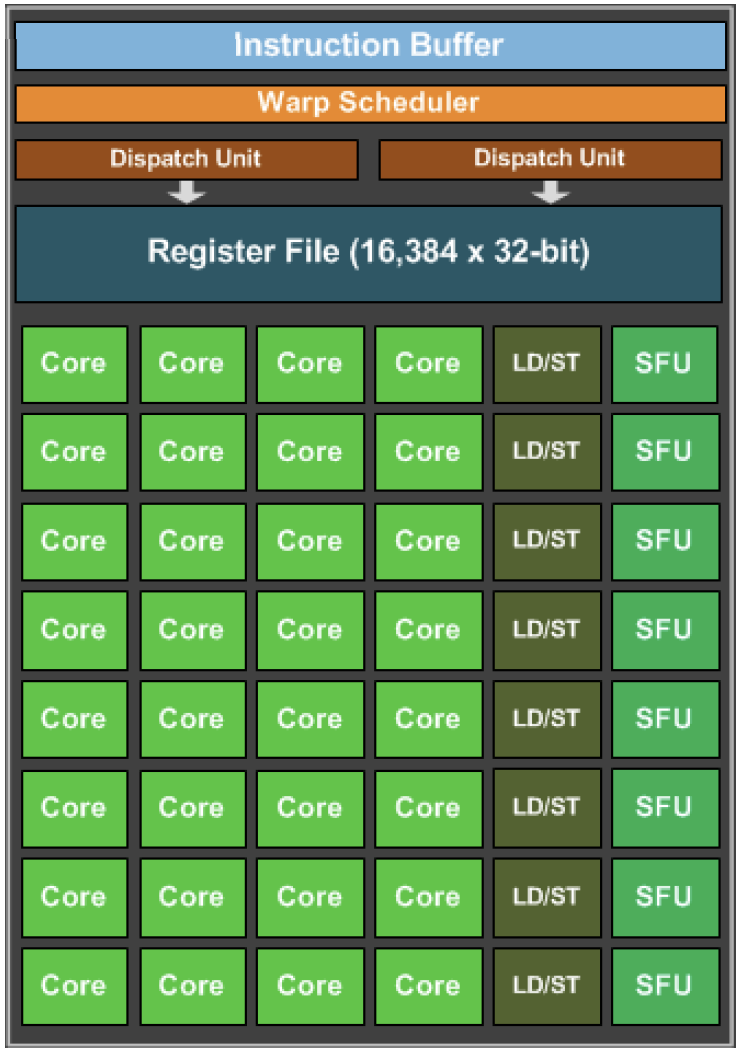


This is the instruction cache
Shared between all cores of the warp.
A dispatch unit issues instructions to a group of cores,
load/store units, or special function units (SFUs).

This is the load/store unit. The hardware component responsible for issuing loads and stores.
Shared between 4 cores.

SFU (The Special Function Unit) is an accelerator to perform tracendental and trigonometric operations such as sine and cosine.
These operations are not executed in parallel; an SFU is shared for those.

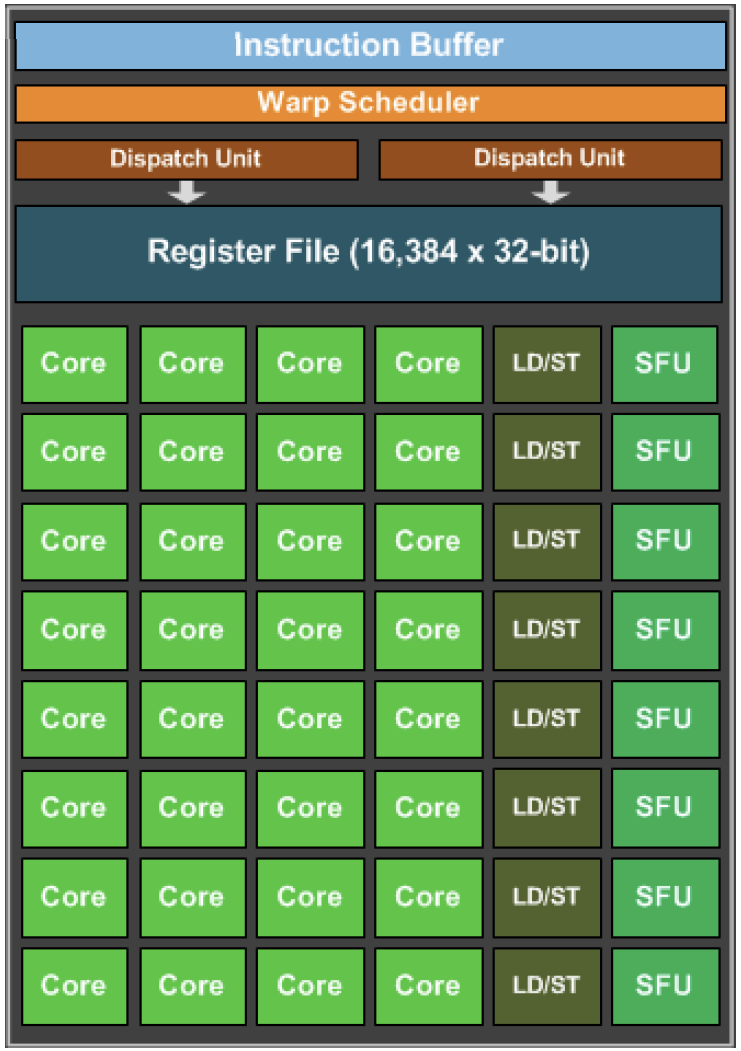
Warp execution



Groups of 32 threads are called a “warp”

They are executed in lock-step, i.e. they all execute the same instruction at the same time

Warp execution



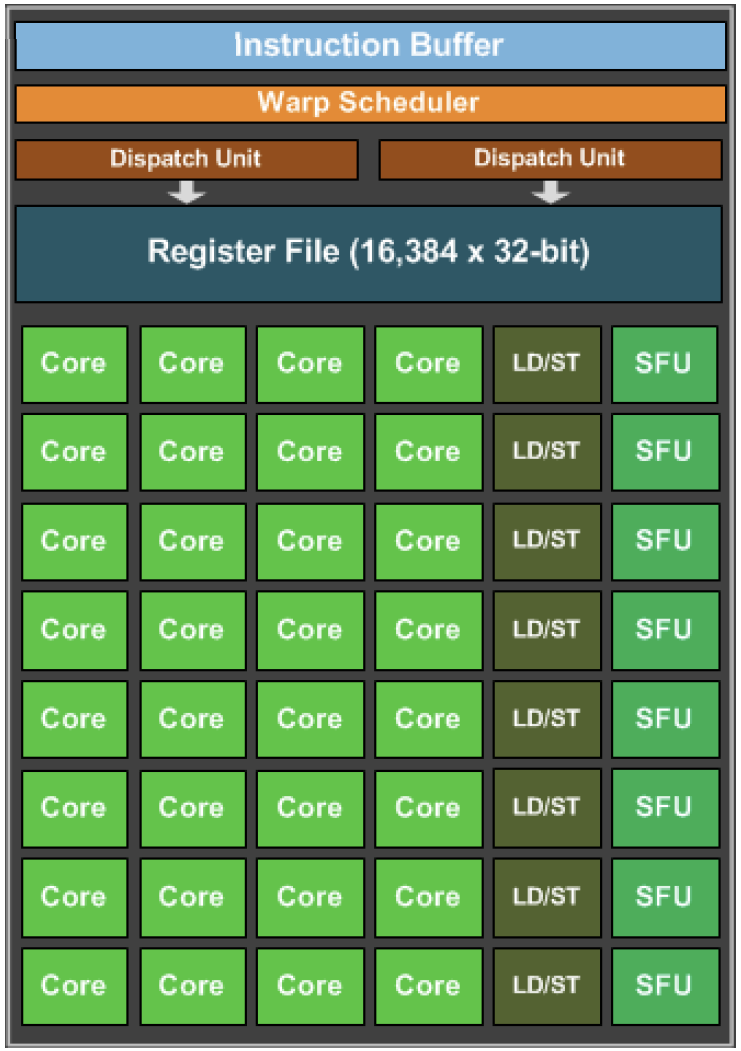
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Program:

```
int variable1 = b[0];
int variable2 = c[0];
int variable3 = variable1 + variable2;
a[0] = variable3;
```

Warp execution



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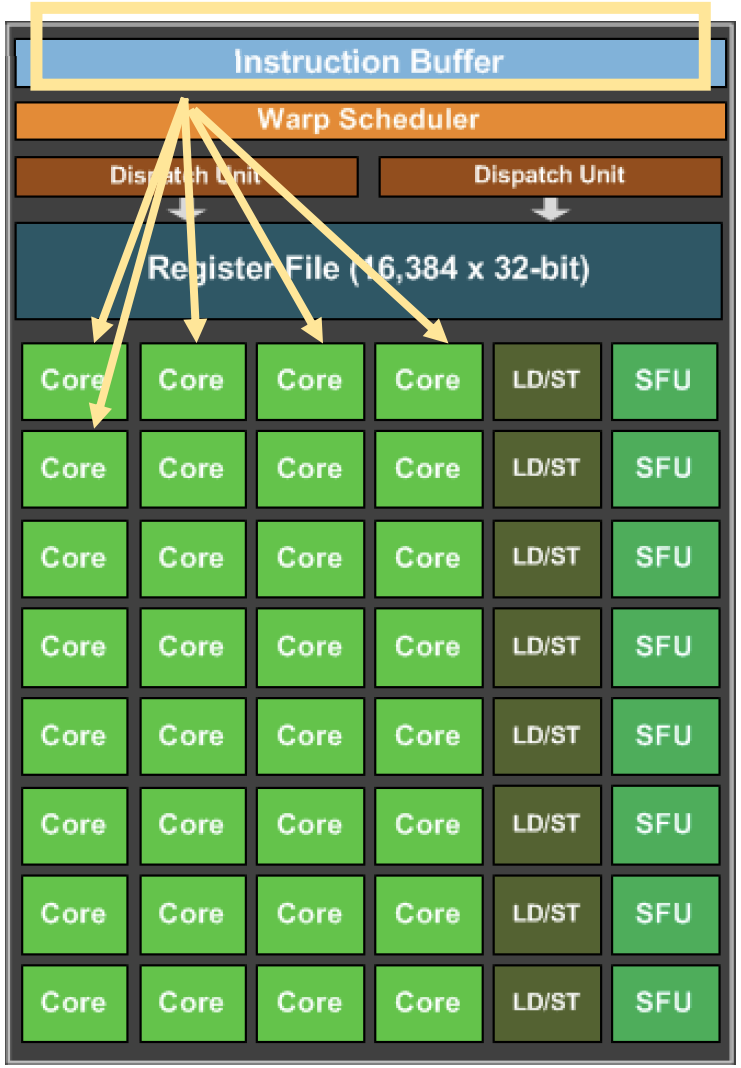
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```

Warp execution

Groups of 32 threads are called a “warp”

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instruction is fetched from the buffer and distributed to all the cores.

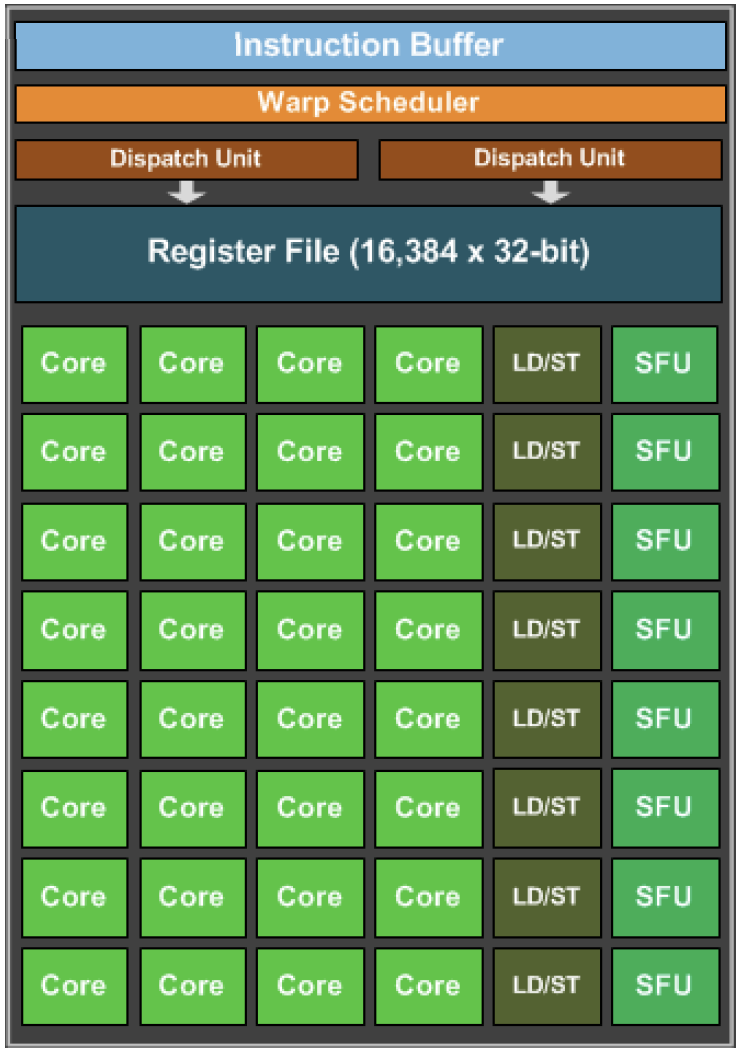
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a[0] = variable3;
```

Warp execution

Groups of 32 threads are called a “warp”

They are executed in lock-step, i.e. they all execute the same instruction at the same time



All cores need to wait until all cores finish the first instruction

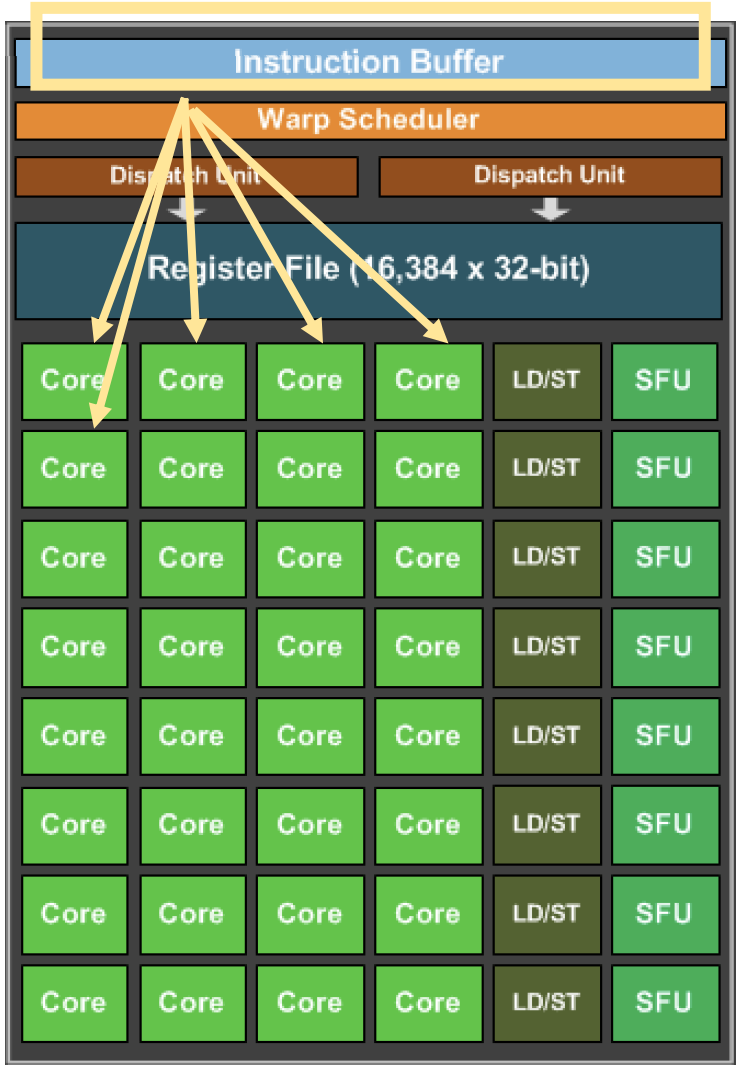
Program:

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a[0] = variable3;
```

Warp execution

Groups of 32 threads are called a “warp”

They are executed in lock-step, i.e. they all execute the same instruction at the same time



Start the next instruction.

Program:

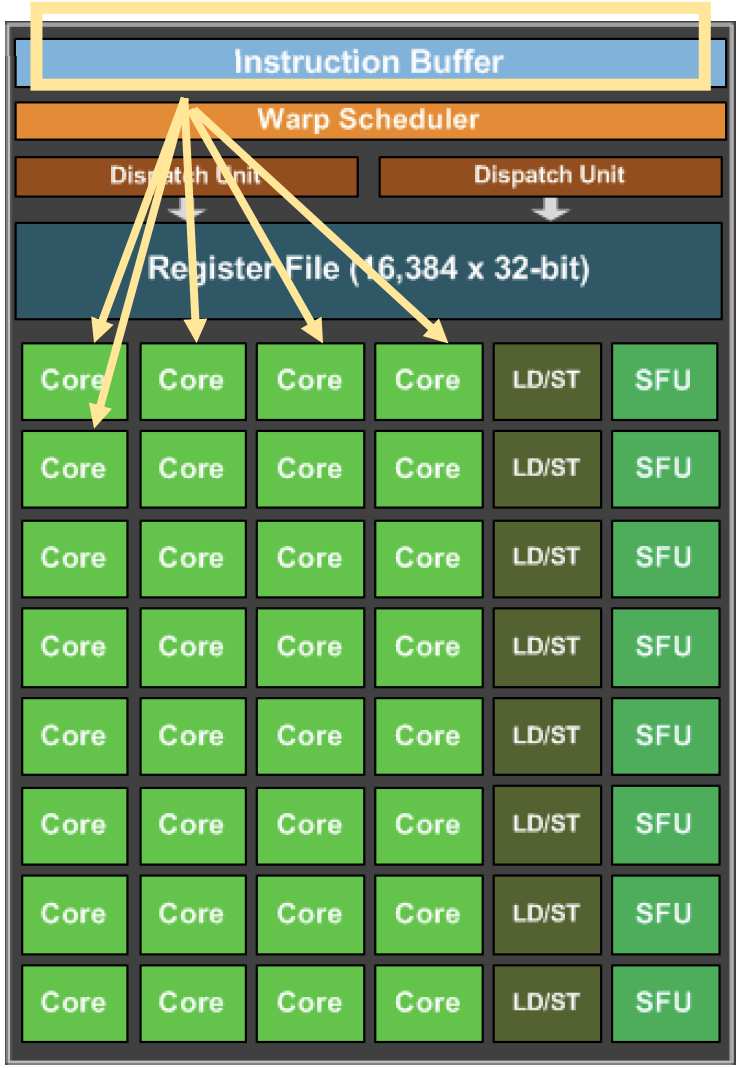
```
int variable1 = b[0];
int variable2 = c[0];
int variable3 = variable1 + variable2;
a[0] = variable3;
```

Why would we have a programming model like this?

Warp execution

Groups of 32 threads are called a “warp”

They are executed in lock-step, i.e. they all execute the same instruction at the same time



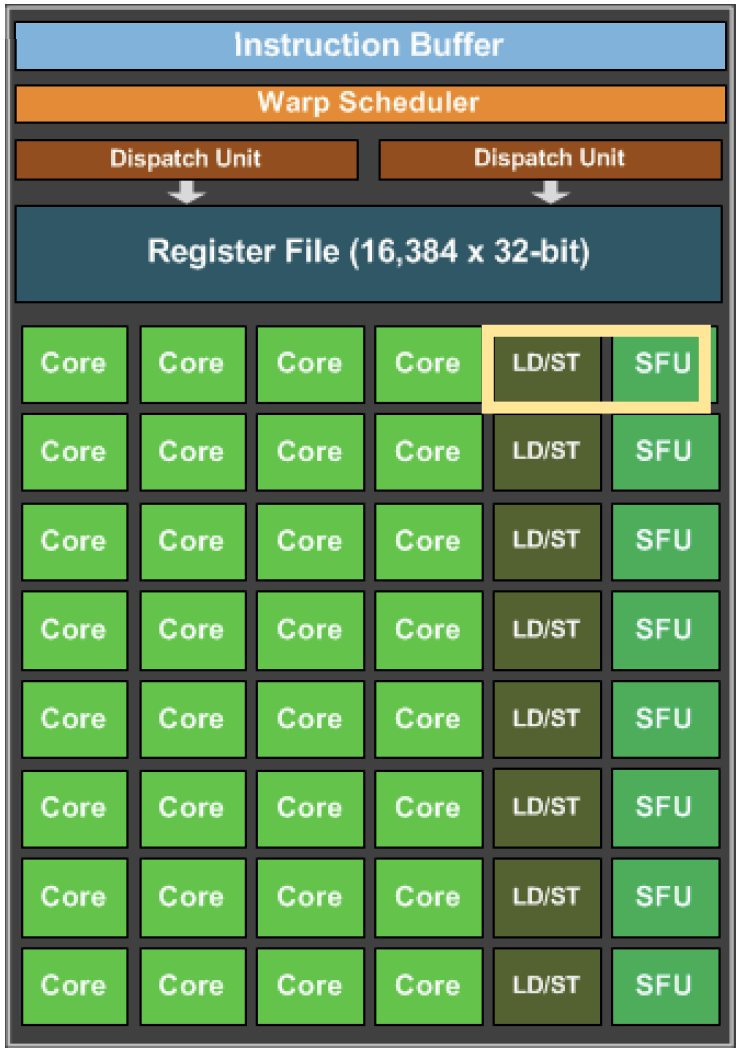
Start the next instruction.

Program:

```
int variable1 = b[0];
int variable2 = c[0];
int variable3 = variable1 + variable2;
a[0] = variable3;
```

Why would we have a programming model like this?
More cores (share program counters)
Can be efficient to share hardware resources

Warp execution



Groups of 32 threads are called a “warp”

They are executed in lock-step, i.e. they all execute the same instruction at the same time

Program:

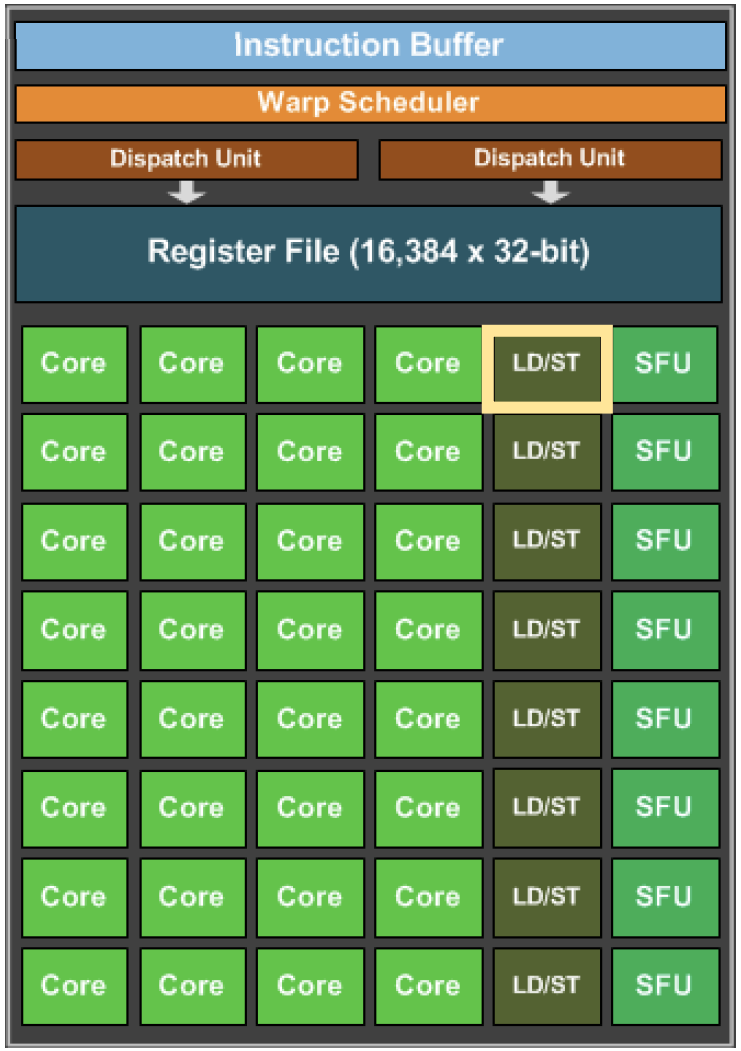
```
int variable1 = b[0];
int variable2 = c[0];
int variable3 = variable1 + variable2;
a[0] = variable3;
```

Why would we have a programming model like this?
More cores (share program counters)
Can be efficient to share hardware resources

Cores share expensive HW units (load/store and special functions)

Warp execution

Lets look closer at memory

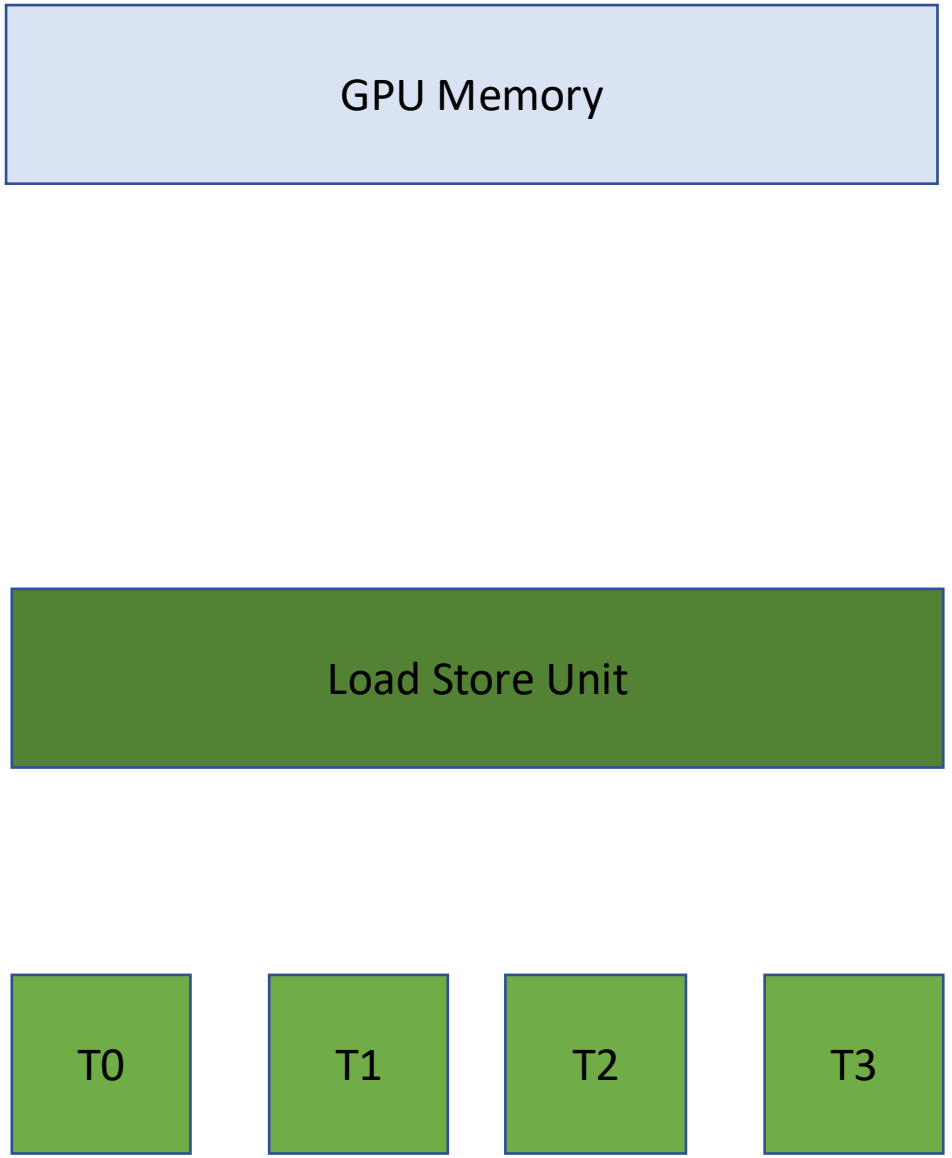


Program:

```
int variable1 = b[0];
int variable2 = c[0];
int variable3 = variable1 + variable2;
a[0] = variable3;
```

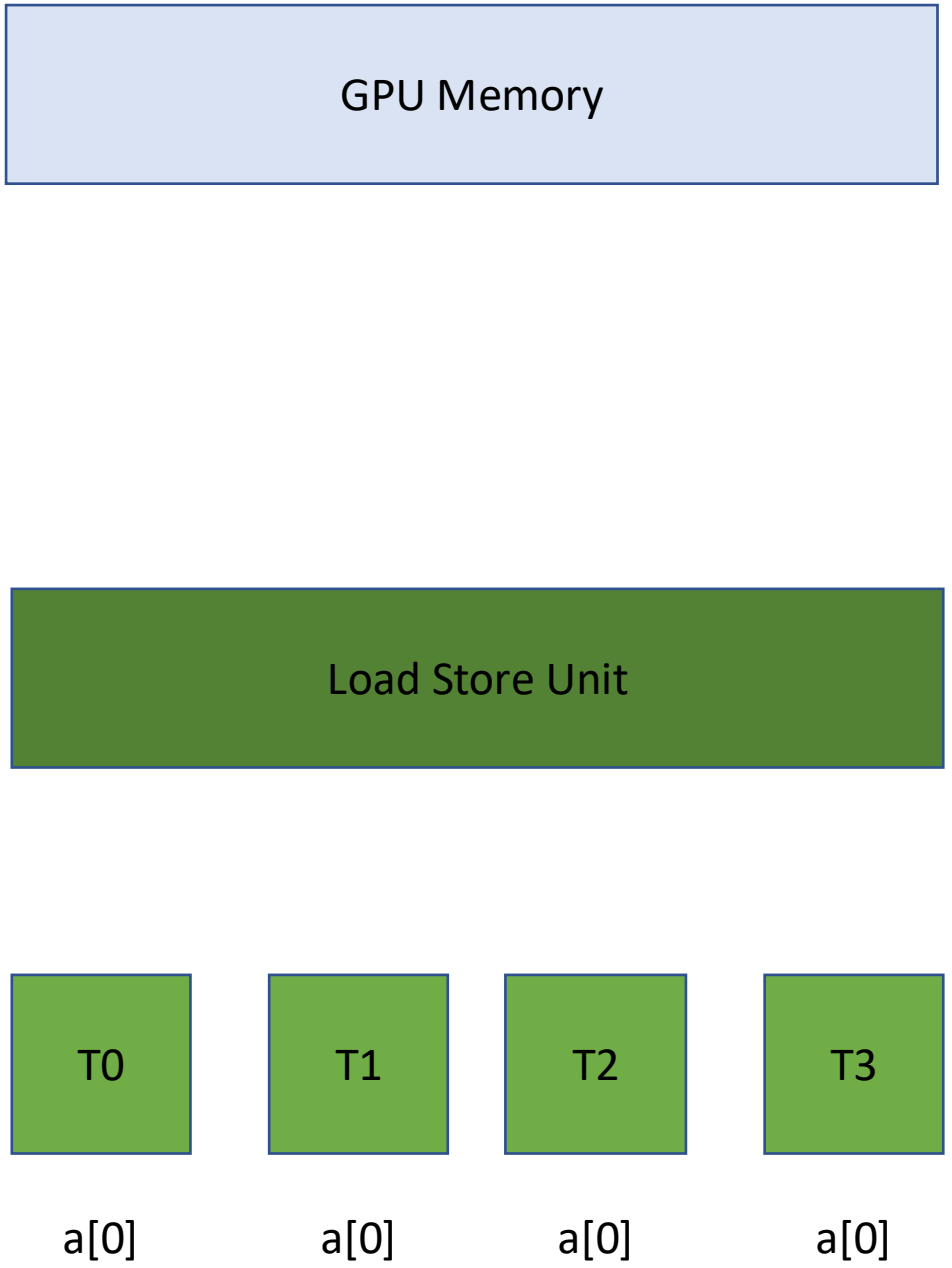
4 cores are accessing memory. what happens if they access the same value?

4 cores are accessing memory. What can happen



4 cores are accessing memory. What can happen

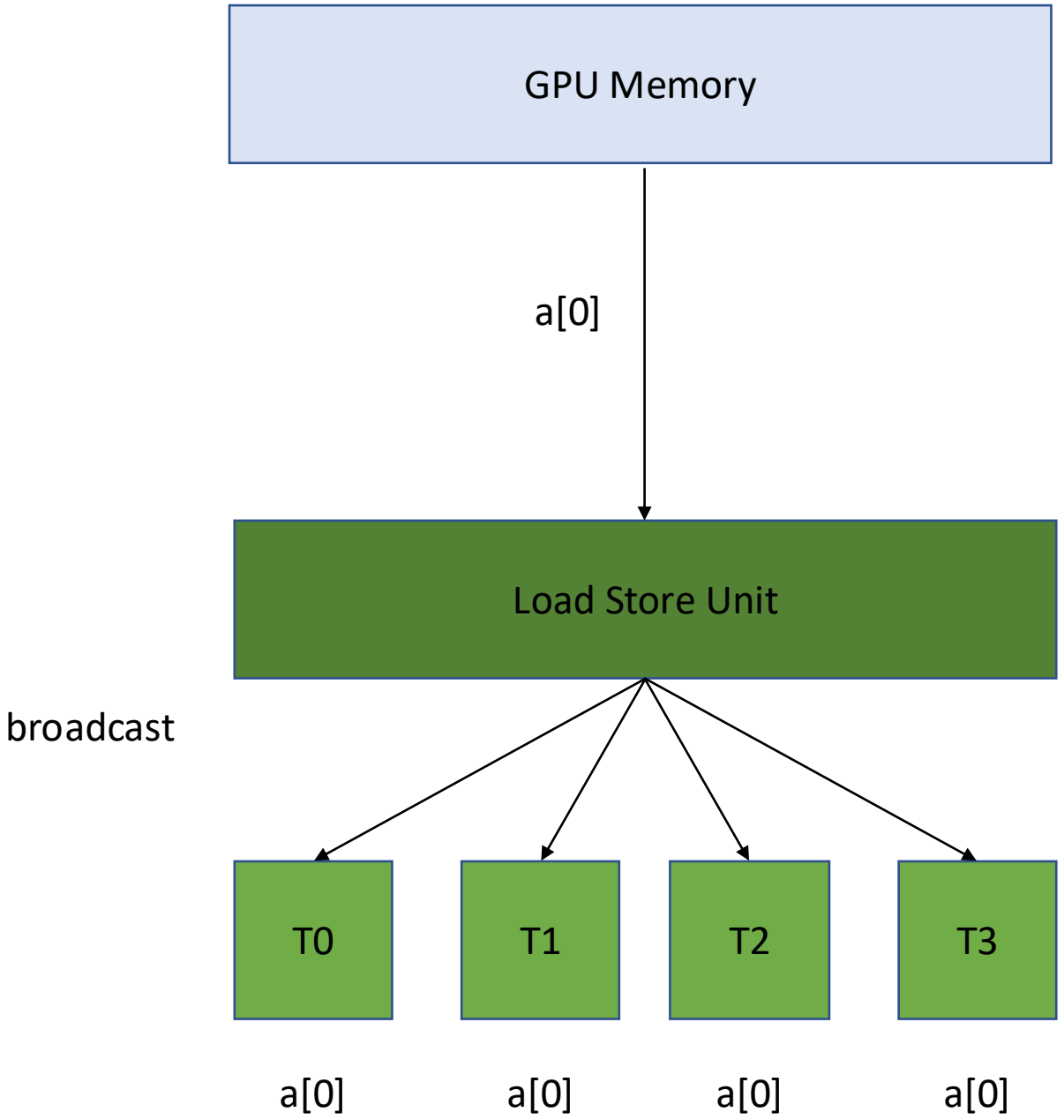
All read the same value



4 cores are accessing memory. What can happen

All read the same value

This is efficient: the load store unit can ask for the value and then broadcast it to all cores.

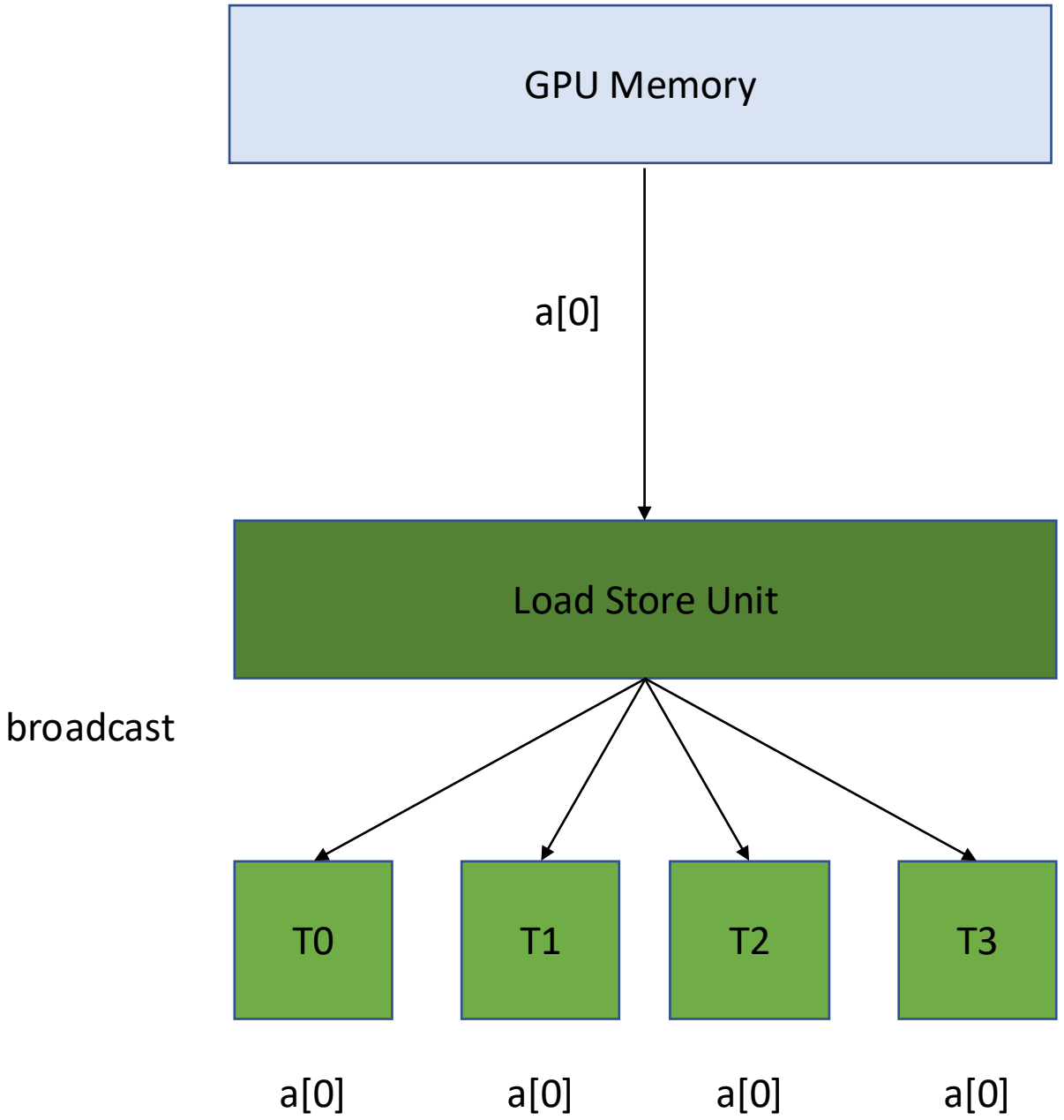


4 cores are accessing memory. What can happen

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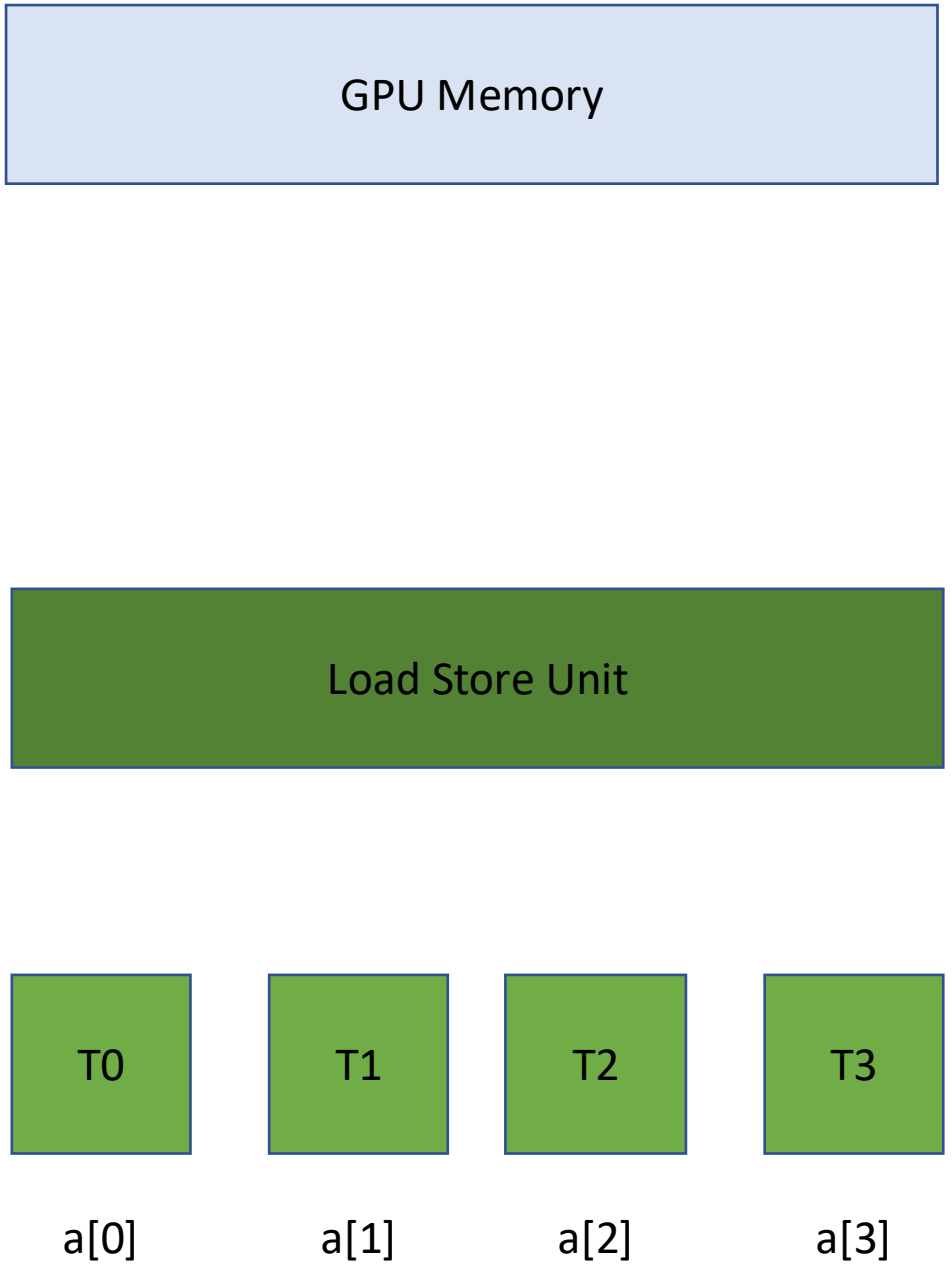
1 request to GPU memory

Efficient, but probably not too common.



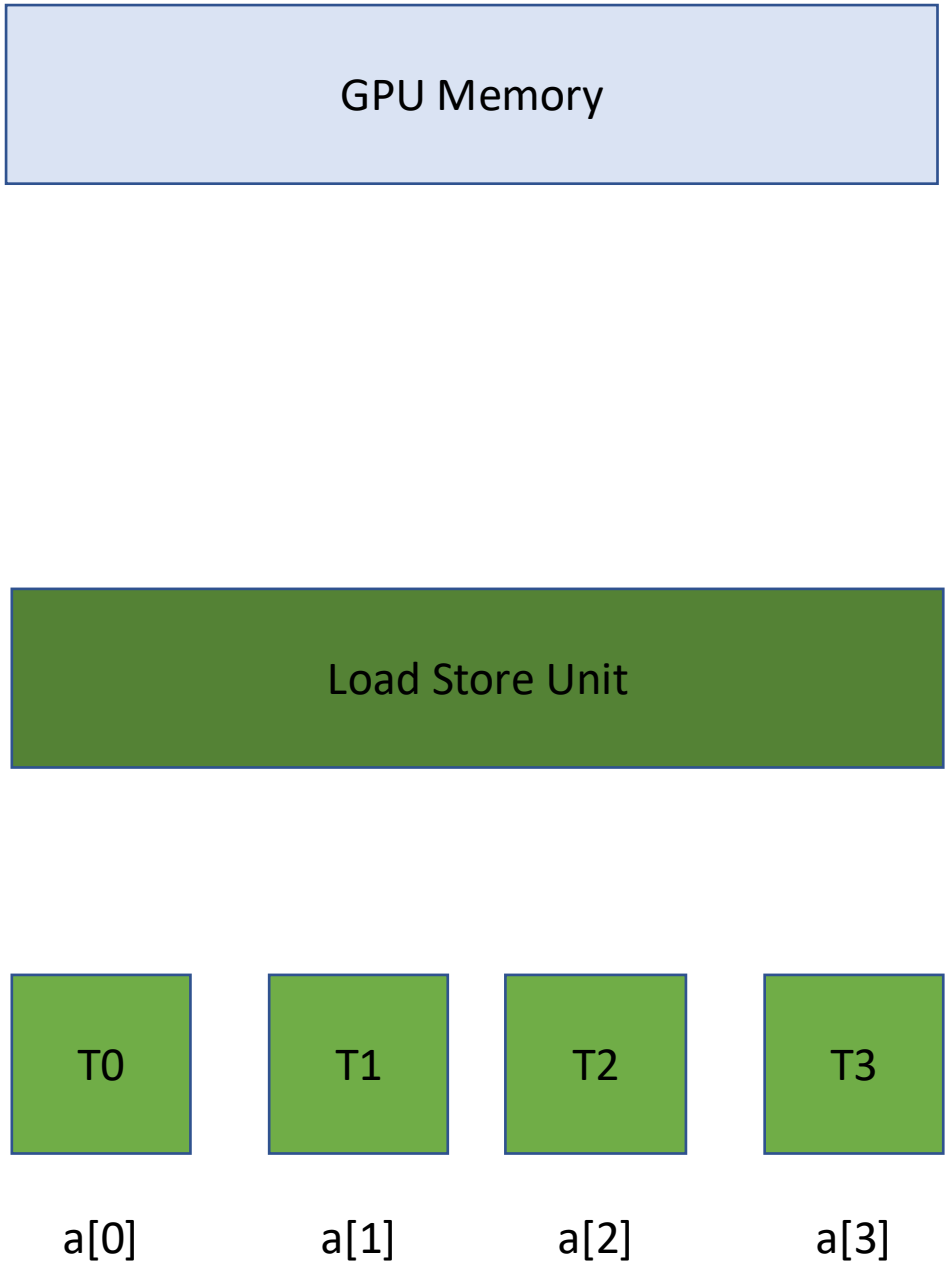
4 cores are accessing memory. What can happen

Read contiguous values



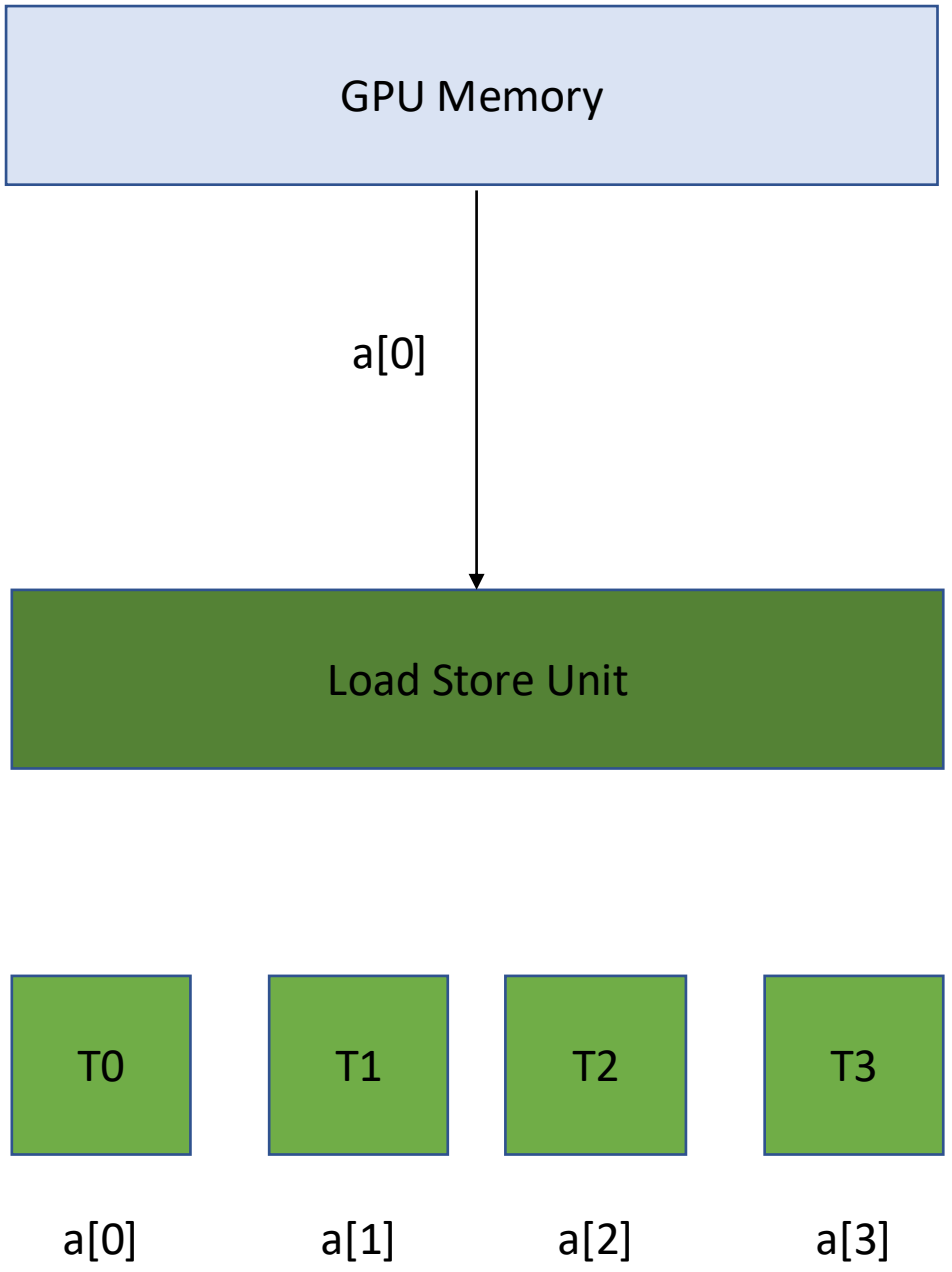
4 cores are accessing memory. What can happen

Read contiguous values
Like the CPU cache, the Load/Store Unit reads in memory in chunks. 16 bytes



4 cores are accessing memory. What can happen

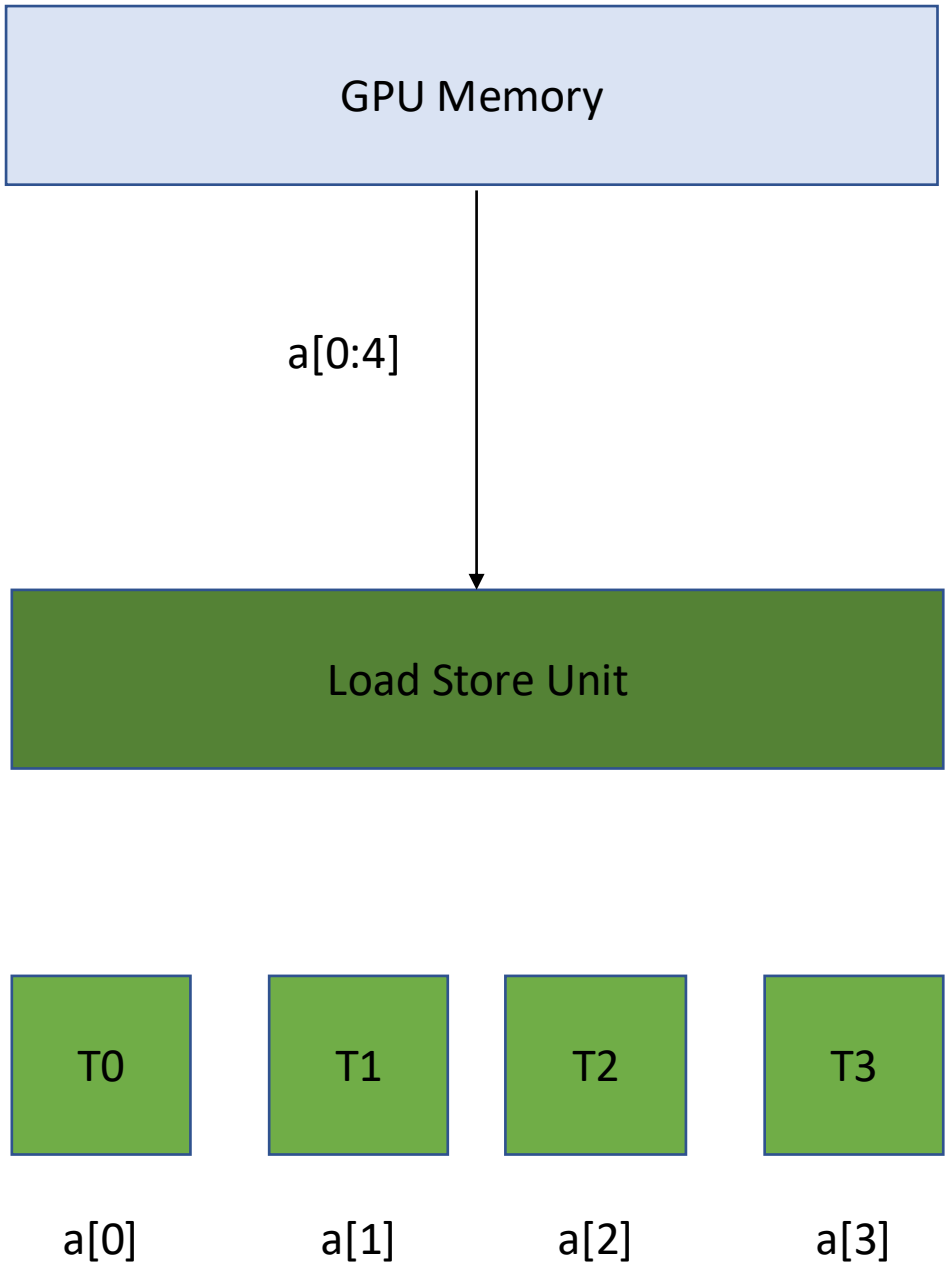
Read contiguous values
Like the CPU cache, the Load/Store Unit reads in memory in chunks. 16 bytes



4 cores are accessing memory. What can happen

Read contiguous values
Like the CPU cache, the Load/Store Unit reads in memory in chunks. 16 bytes

Can easily distribute the values to the threads

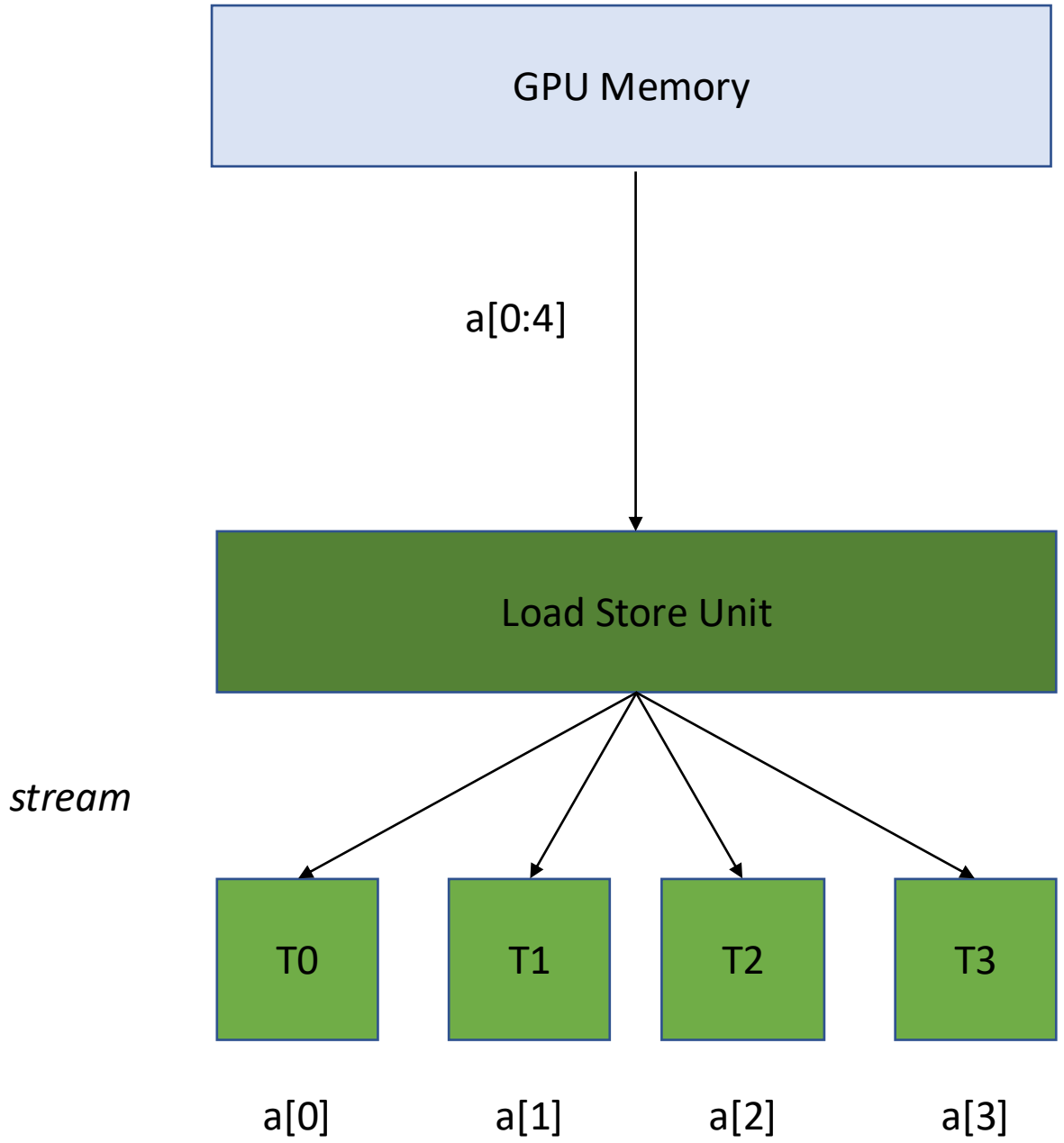


4 cores are accessing memory. What can happen

Read contiguous values
Like the CPU cache, the Load/Store Unit reads in memory in chunks. 16 bytes

Can easily distribute the values to the threads

1 request to GPU memory

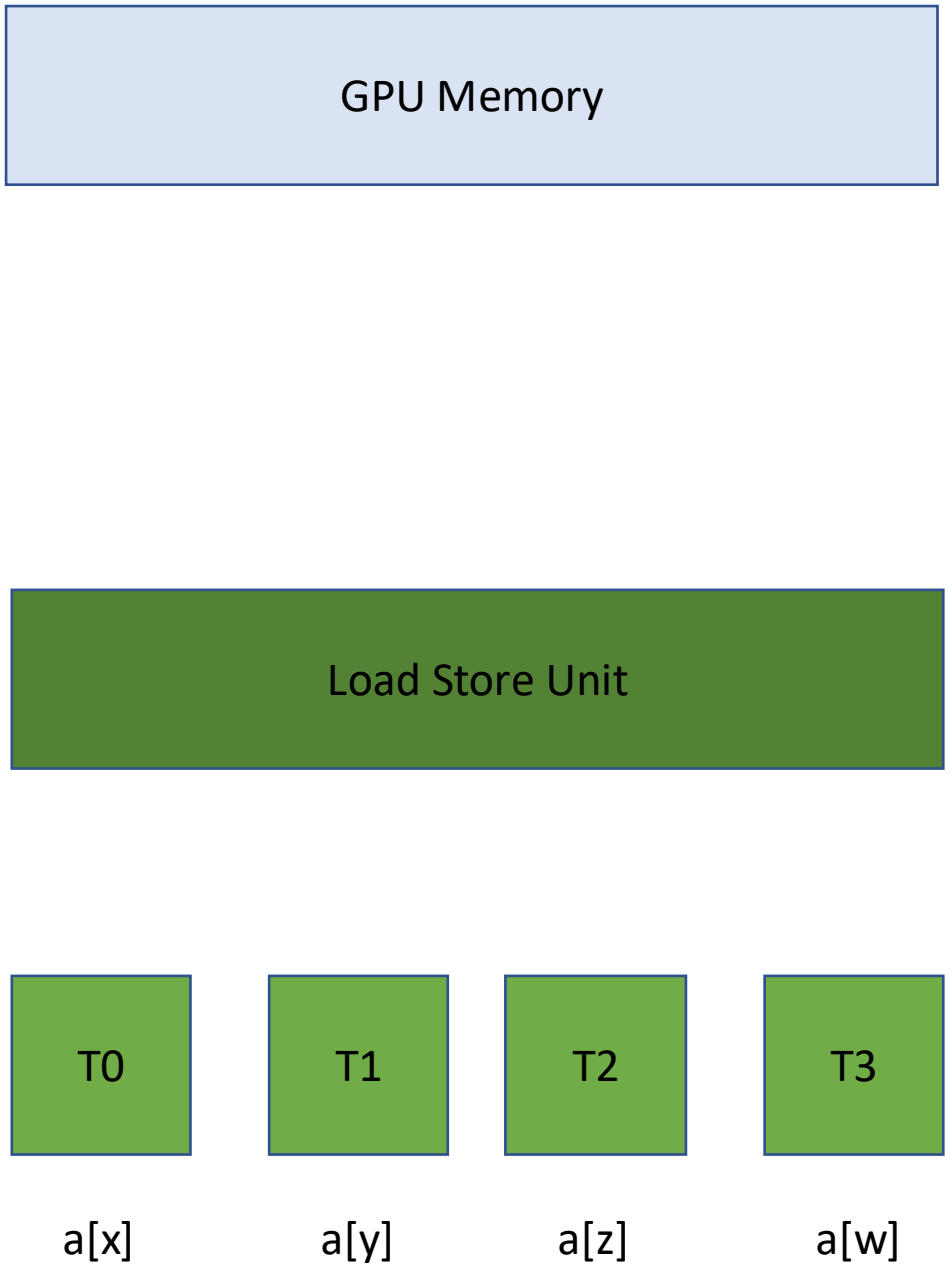


4 cores are accessing memory. What can happen

Read non-contiguous values

Not good!

Accesses are Serialized.
You need 4 requests to GPU memory

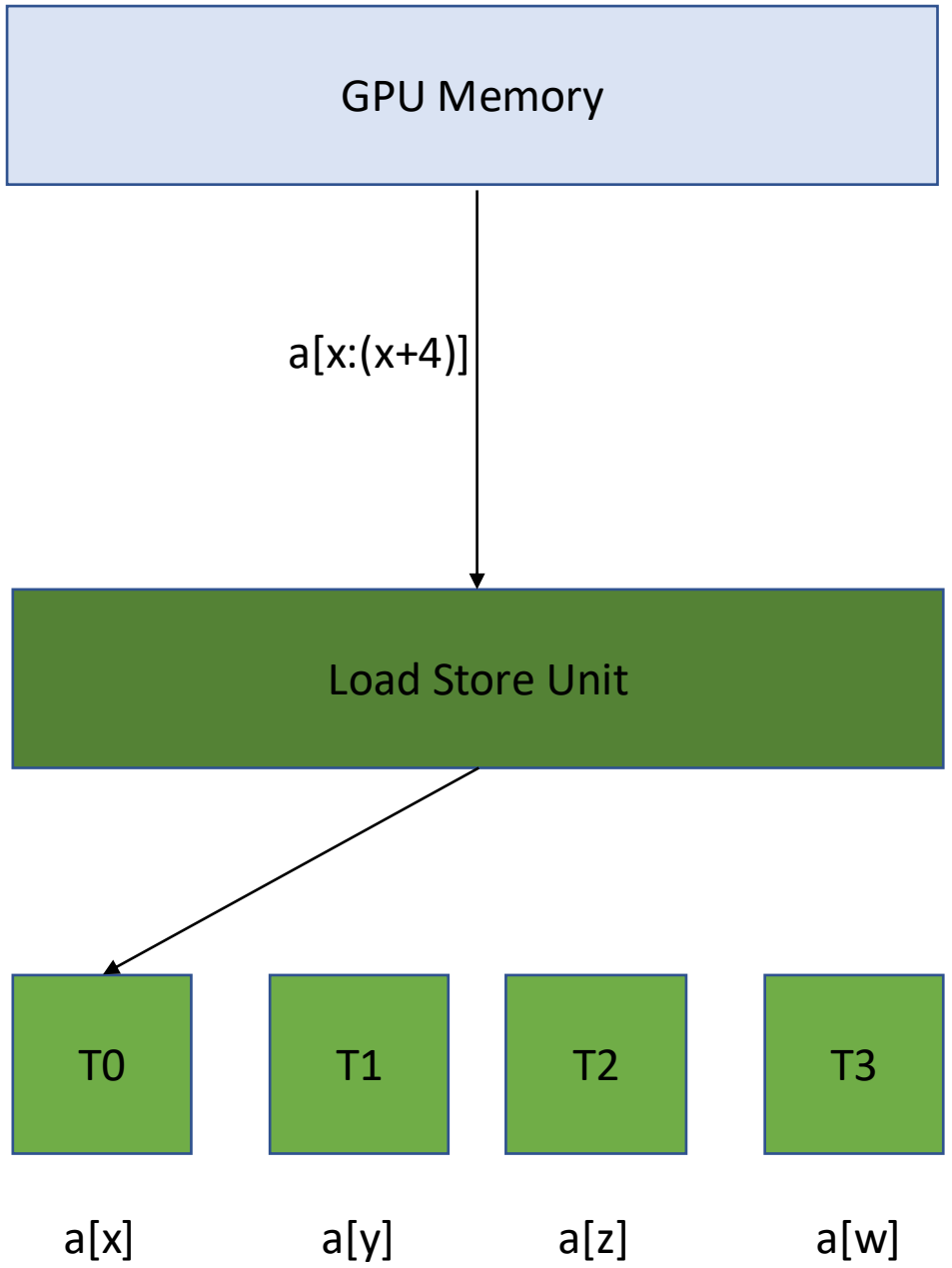


4 cores are accessing memory. What can happen

Read non-contiguous values

Not good!

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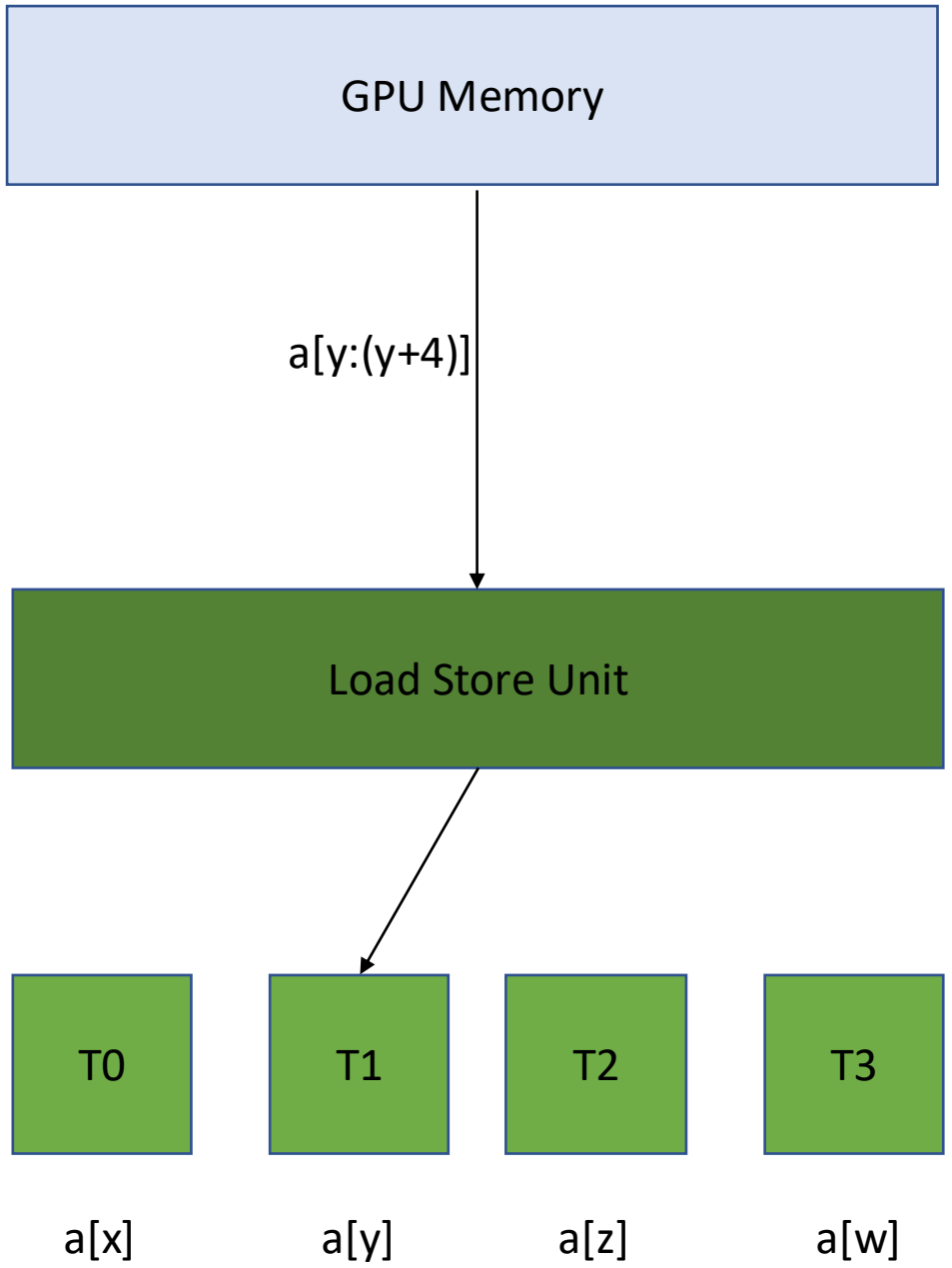


4 cores are accessing memory. What can happen

Read non-contiguous values

Not good!

Accesses are Serialized.
You need 4 requests to GPU memory

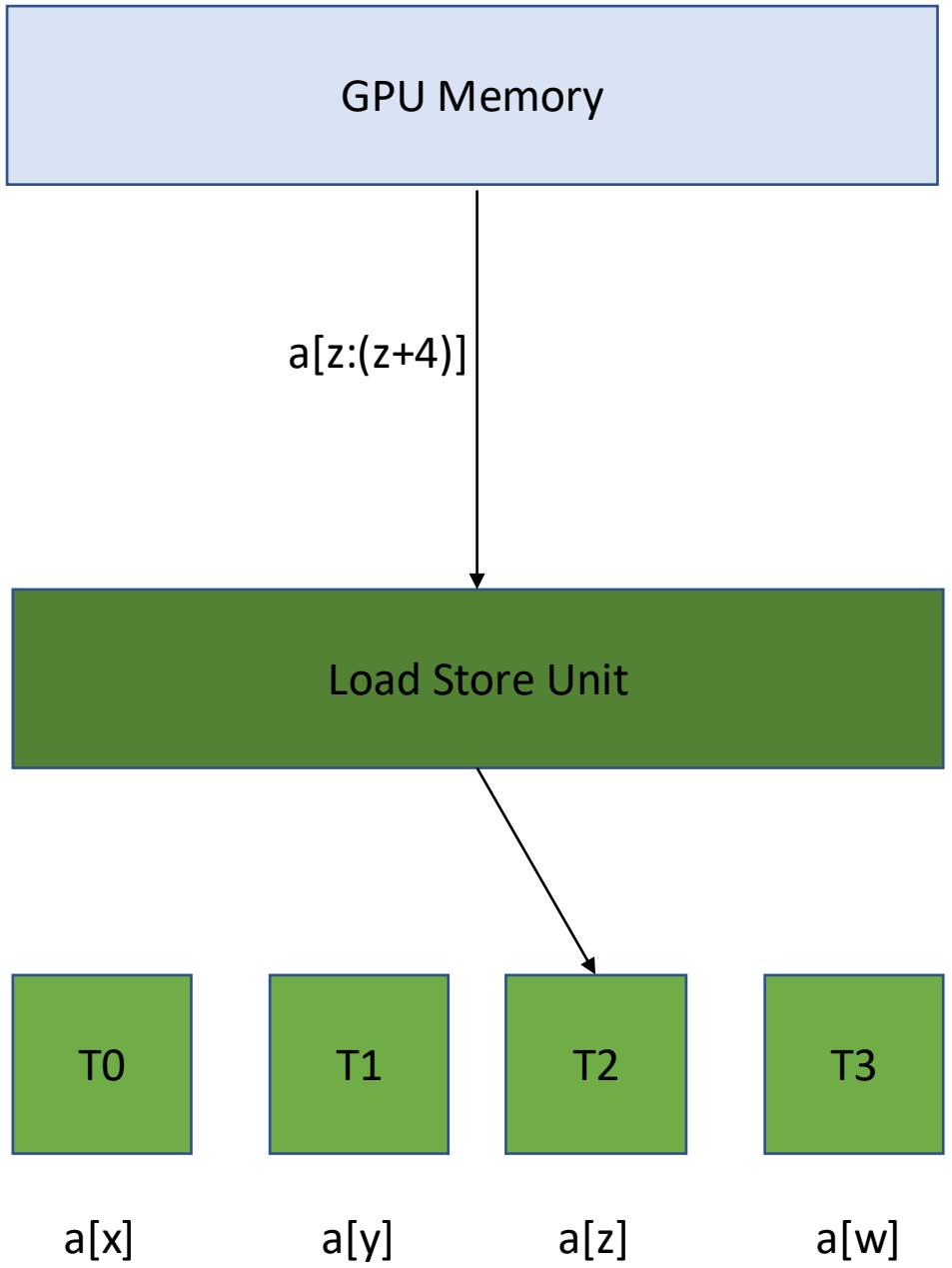


4 cores are accessing memory. What can happen

Read non-contiguous values

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Accesses are Serialized.
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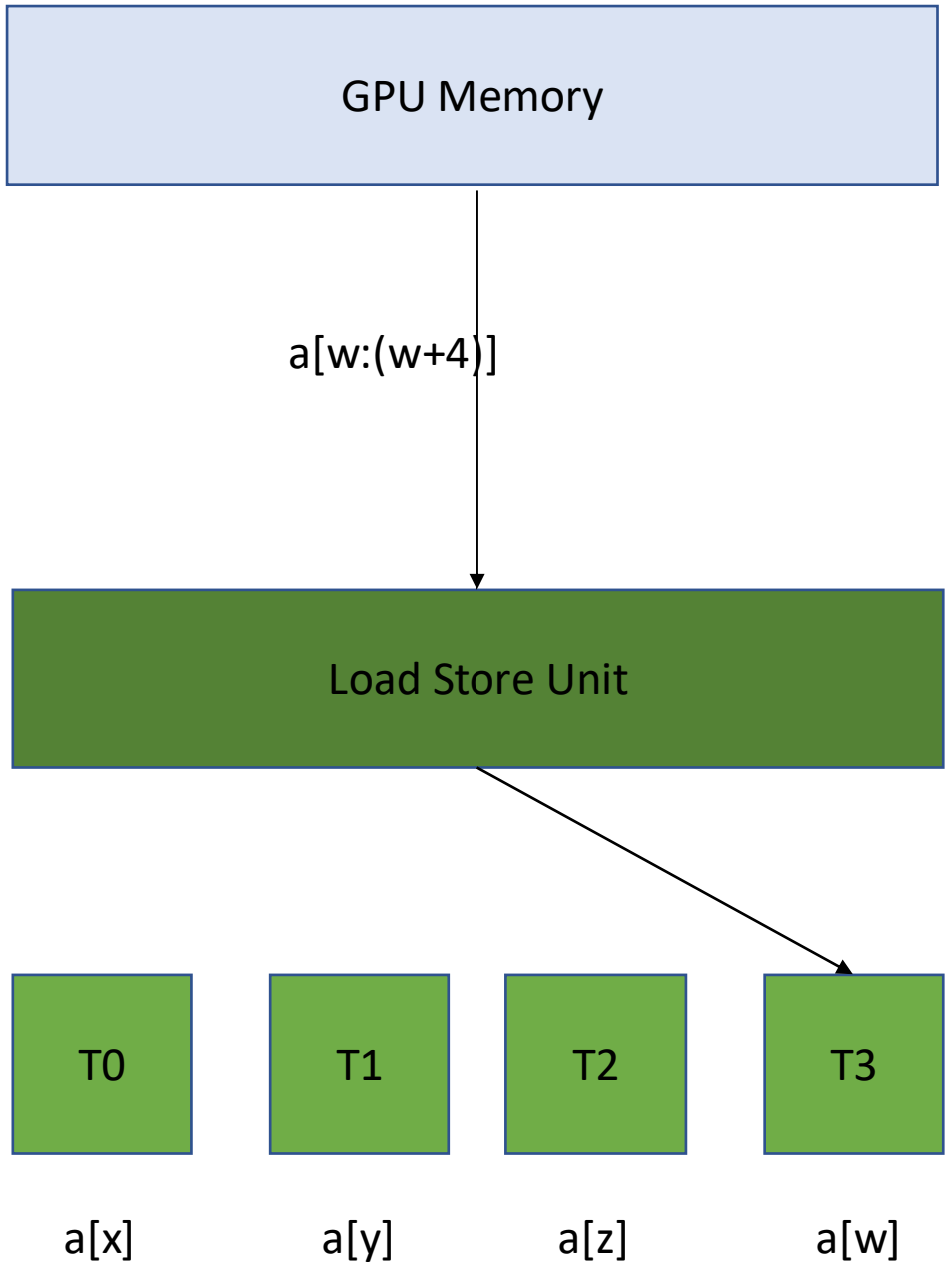


4 cores are accessing memory. What can happen

Read non-contiguous values

Not good!

Accesses are Serialized.
You need 4 requests to GPU memory



Quiz 12.

Like the CPU cache, the Load/Store Unit reads in memory in chunks. Is this correct?

Select one

A. True

B. False

Scan to answer

forms.gle/c8hpnGUtrui63zBcA

chart: docs.google.com/forms/d/e/1FAIpQLSc53ivr_o0JWDVCwQwmlR4_7xFgn_3t4vDOA2QgQjEa9rBsbQ/viewanalytics

Access pattern

```
__global__ void vector_add(int * d_a, int * d_b, int * d_c, int size) {
    int chunk_size = size/blockDim.x;
    int start = chunk_size * threadIdx.x;
    int end = start + chunk_size;
    for (int i = start; i < end; i++) {
      d_a[i] = d_b[i] + d_c[i];
   }
}
```

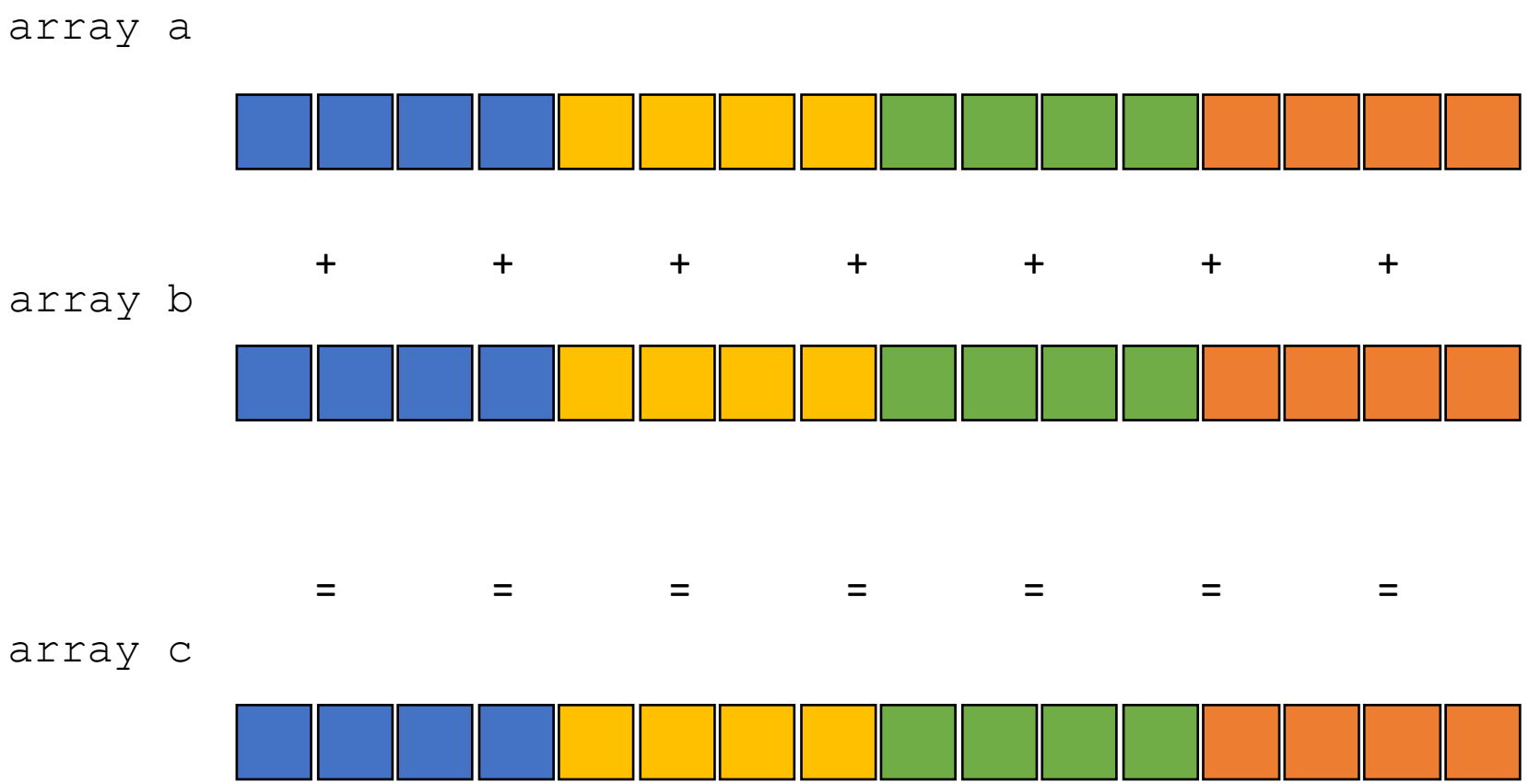
calling the function

```
vector_add<<<1,1024>>>(d_a, d_b, d_c, size);
```

Chunked Pattern

Computation can easily be divided into threads

Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange

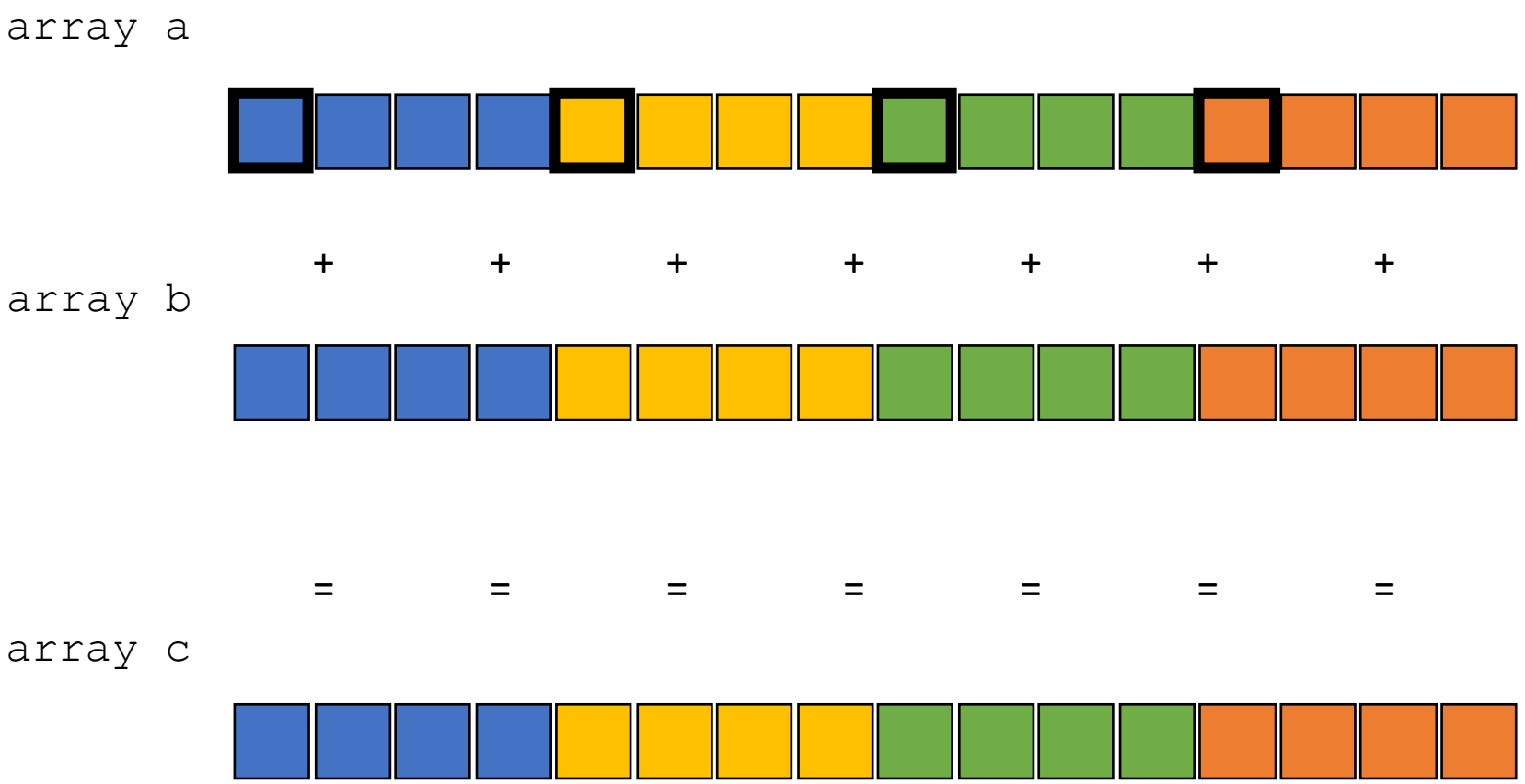


Chunked Pattern

the first element accessed by the 4 threads sharing a load store unit. What sort of access is this?

Computation can easily be divided into threads

Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange



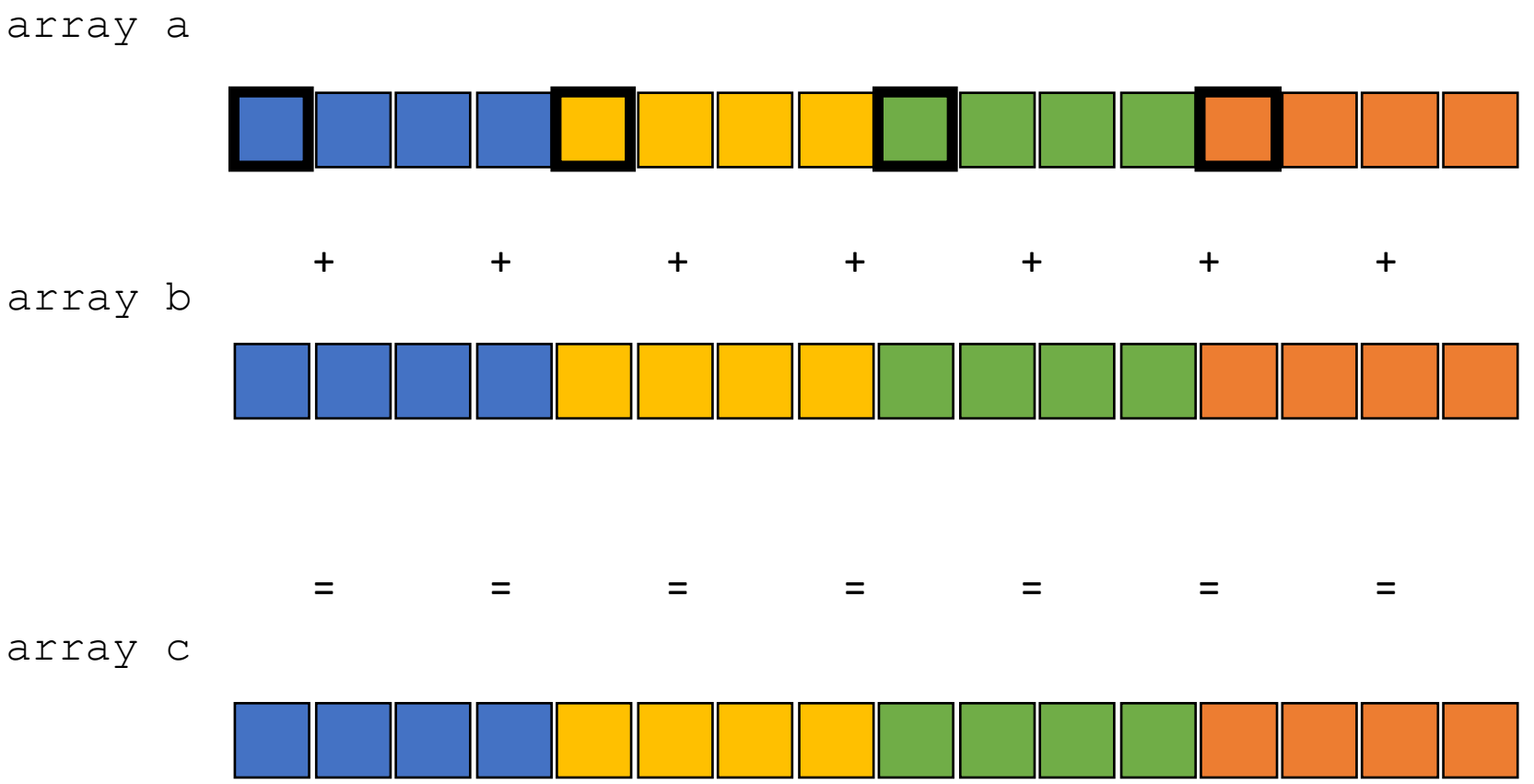
Chunked Pattern

the first element accessed by the 4 threads sharing a load store unit. What sort of access is this?

How can we fix this?

Computation can easily be divided into threads

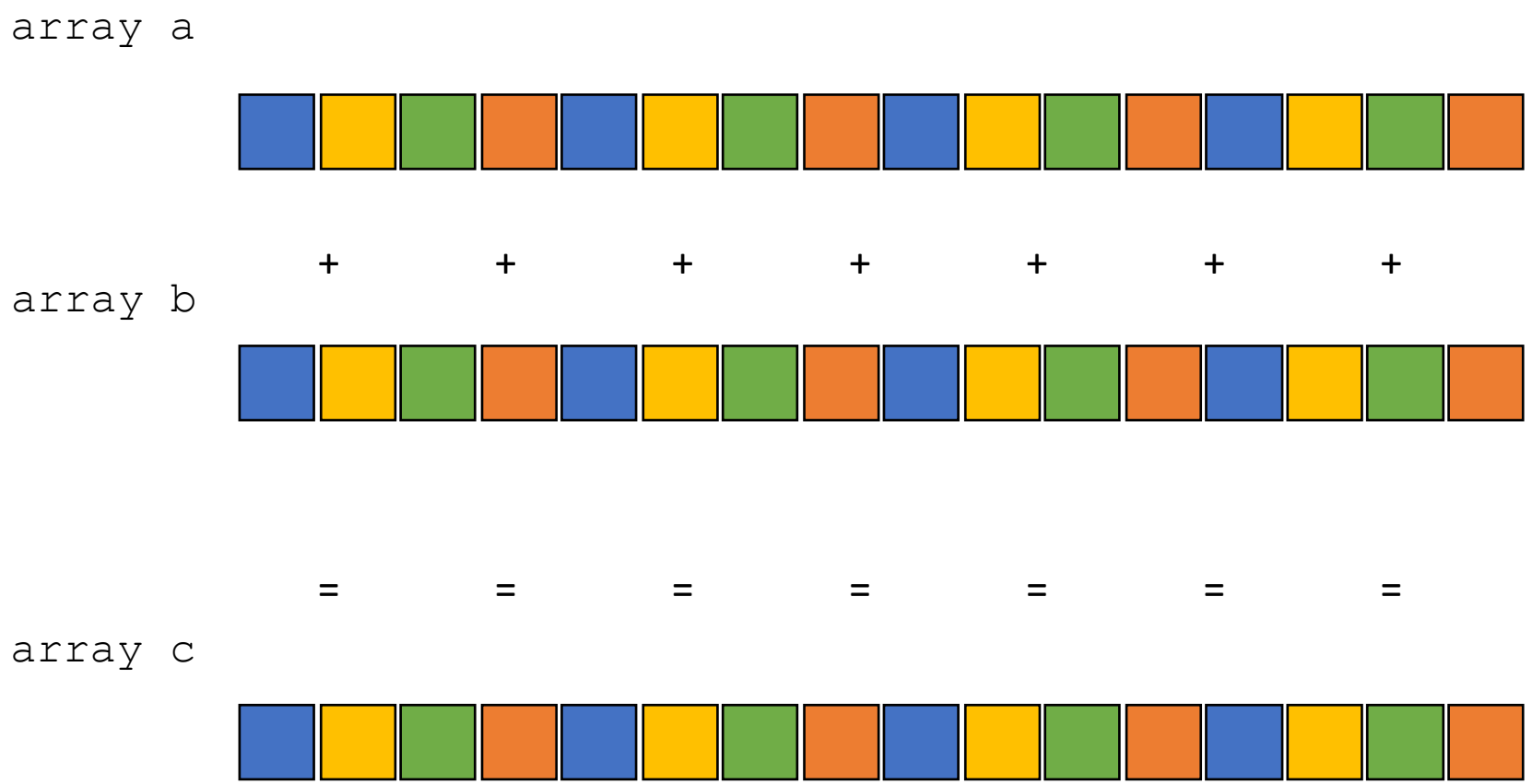
Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange



Stride Pattern

Computation can easily be divided into threads

Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange

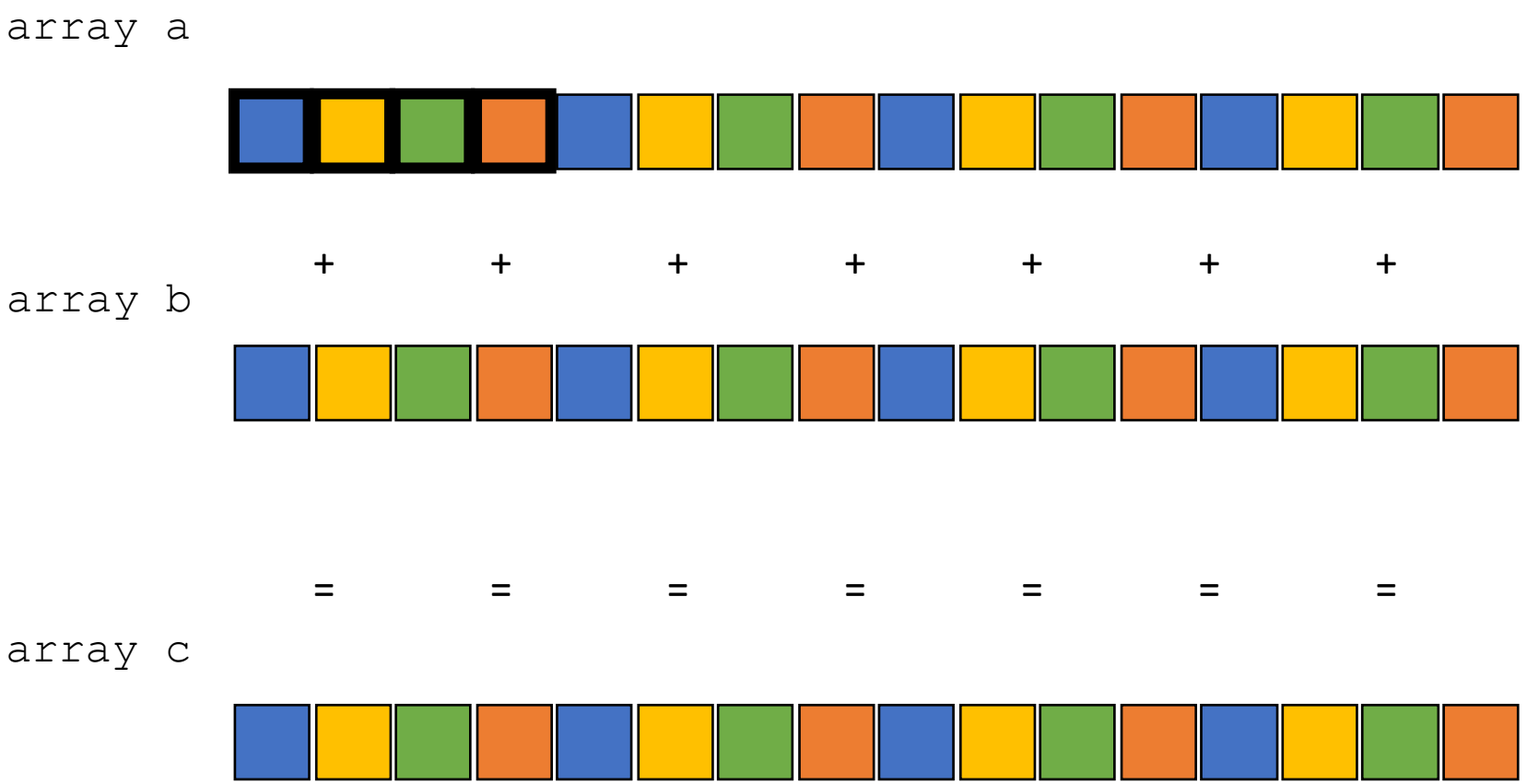


Stride Pattern

What sort of pattern is this?

Computation can easily be divided into threads

Thread 0 - Blue
Thread 1 - Yellow
Thread 2 - Green
Thread 3 - Orange



Go back to our program

```
__global__ void vector_add(int * d_a, int * d_b, int * d_c, int size) {
    int chunk_size = size/blockDim.x;
    int start = chunk_size * threadIdx.x;
    int end = start + chunk_size;
    for (int i = start; i < end; i++) {
      d_a[i] = d_b[i] + d_c[i];
   }
}
```

Chunked pattern
Let's change this to a stride pattern

calling the function

```
vector_add<<<1,1024>>>(d_a, d_b, d_c, size);
```

blockIdx.x is the block id.
blockDim.x is the number of threads per block.
threadIdx.x is the thread id within block.

Go back to our program

```
__global__ void vector_add(int * d_a, int * d_b, int * d_c, int size) {
  for (int i = threadIdx.x; i < size; i+=blockDim.x) {
     d_a[i] = d_b[i] + d_c[i];
  }
}
```

calling the function

```
vector_add<<<1,1024>>>(d_a, d_b, d_c, size);
```

blockIdx.x is the block id.
blockDim.x is the number of threads per block.
threadIdx.x is the thread id within block.

Coalesced memory accesses

This is about 4x faster.

Quiz 12.

What can accelerate memory accesses on GPUs?

Free response: a few sentences on the form



Scan to answer

forms.gle/u2GrZU9t4BPo1iES9

chart: docs.google.com/forms/d/e/1FAIpQLSfqLApVqPTdL8tkgO51FhYpmg-18blEHVSUIyKfQ_NwODqnow/viewanalytics

Quiz 12.

What can accelerate memory accesses on GPUs?

Free response: a few sentences on the form

Memory coalescing accelerates execution.

Scan to answer

forms.gle/u2GrZU9t4BPo1iES9

chart: docs.google.com/forms/d/e/1FAIpQLSfqLApVqPTdL8tkgO51FhYpmg-18blEHVSUIyKfQ_NwODqnow/viewanalytics

Coalesced memory accesses

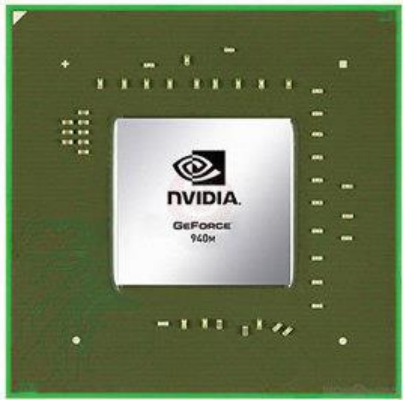
This is about 4x faster.



What else can we do?

Final Round

Tiny GPU in an embedded system



Fight!



The CPU in my professor workstation



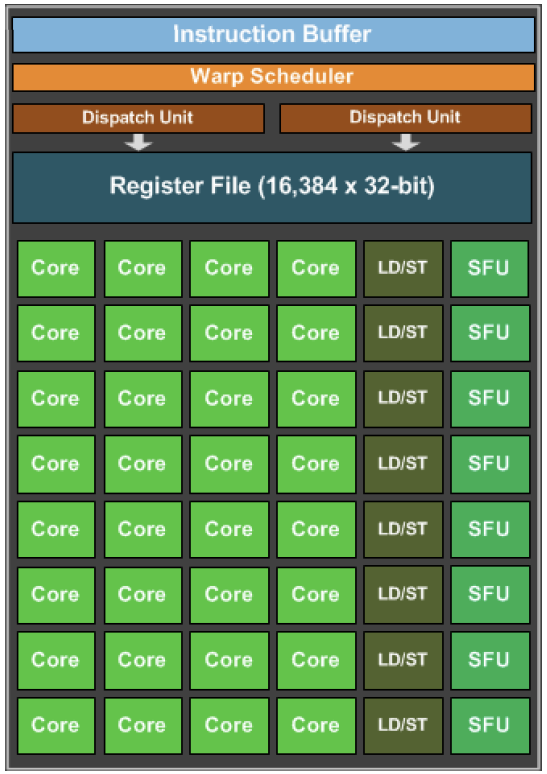
Nvidia Jetson Nano (whole chip, CPU + GPU)
2 Billion transistors
10 TDP
Est. $99

Intel i7-9700K
2.16 Billion transistors
95 TDP
Est. $316

https://www.techpowerup.com/gpu-specs/geforce-940m.c2648
https://www.alibaba.com/product-detail/Intel-Core-i7-9700K-8-Cores_62512430487.html
https://www.prolast.com/prolast-elevated-boxing-rings-22-x-22/

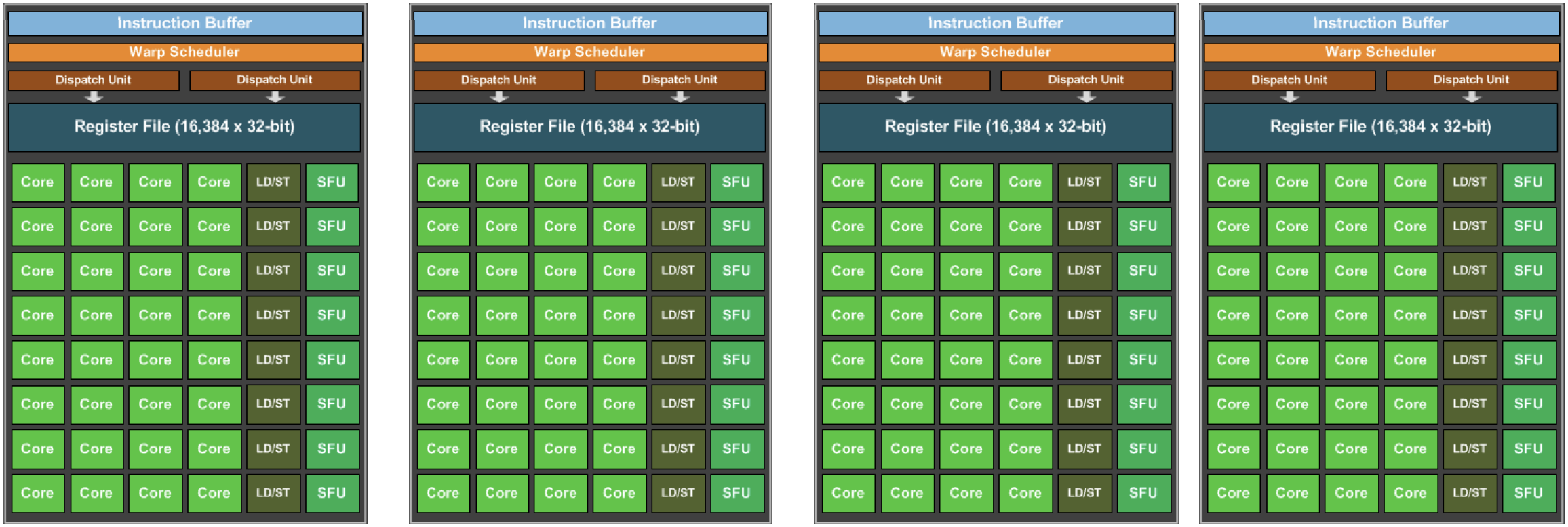
Multiple streaming multiprocessors

We've been talking only about 1 streaming multiprocessor, most GPUs have multiple SMs big ML GPUs have 32.



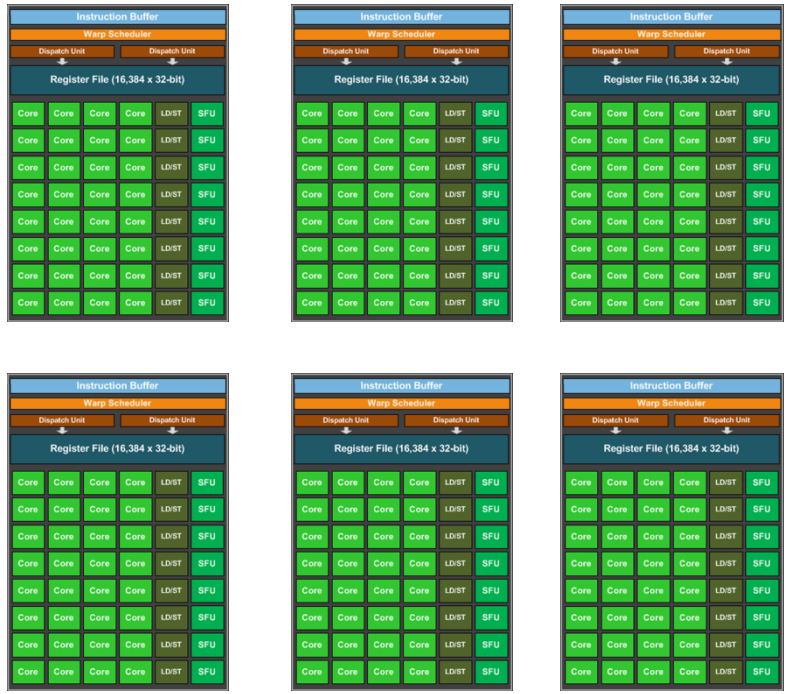
Multiple streaming multiprocessors

We've been talking only about 1 streaming multiprocessor, most GPUs have multiple SMs big ML GPUs have 32. This little GPU has 4



Multiple streaming multiprocessors

CUDA provides virtual streaming multiprocessors called **blocks**
They hide memory latency
Very efficient at launching and joining **blocks.**
No limit on blocks: launch as many as you need to map 1 thread to 1 data element



Go back to our program

```
__global__ void vector_add(int * d_a, int * d_b, int * d_c, int size) {
  for (int i = threadIdx.x; i < size; i+=blockDim.x) {
     d_a[i] = d_b[i] + d_c[i];
  }
}
```

calling the function

```
vector_add<<<1,1024>>>(d_a, d_b, d_c, size);
```

Launch with many thread blocks

blockIdx.x is the block id.
blockDim.x is the number of threads per block.
threadIdx.x is the thread id within block.

Go back to our program

```
__global__ void vector_add(int * d_a, int * d_b, int * d_c, int size) {
  int i = blockIdx.x * blockDim.x + threadIdx.x;
  d_a[i] = d_b[i] + d_c[i];
}
```

calling the function

```
#define SIZE (1024*1024)
vector_add<<<1024,1024>>>(d_a, d_b, d_c, SIZE);
```

Launch with many thread blocks

Need to recalculate some thread ids.

Now we have 1 thread for each element

blockIdx.x is the block id.
blockDim.x is the number of threads per block.
threadIdx.x is the thread id within block.

Quiz 12. Q4

Why do we need to calculate int i = blockIdx.x * blockDim.x + threadIdx.x?

Select one

A. To get the index in the matrix we want to compute in that thread

B. To get the whole matrix we want to compute in that thread

C. To get the index in the matrix we want to compute in that warp



Scan to answer

forms.gle/nCPhnjB89Kb1iF8t6

chart: docs.google.com/forms/d/e/1FAIpQLSdPYUn4_ybwmH3S_r8JaguP6oDhihRmp0GREkT-t_5T_nlGKA/viewanalytics

Final round

With 10x energy efficiency, GPU is as fast as CPU.

A datacenter GPU: an H100 at 700 W.
With the same energy efficiency, GPU is about 33x as CPU.

2D or 3D

```
dim3 block(32, 32);                              // 1024 threads per block
                                                 // defaulting to 1
dim3 grid((W + 31) / 32, (H + 31) / 32);         // enough blocks to cover W x H
                                                 // A grid is all the blocks created
                                                 //   by one kernel launch.
blur<<<grid, block>>>(in, out, W, H);

__global__ void blur(float *in, float *out, int W, int H) {
  int x = blockIdx.x * blockDim.x + threadIdx.x;
  int y = blockIdx.y * blockDim.y + threadIdx.y;
  if (x < W && y < H)
    out[y * W + x] = ...;
}
```

thread, warp (32 threads, scheduled as a unit), block (up to 1024 threads, 32 warps on one SM), grid.

Consequences of Warps, Stride Instructions

```
__global__ void vector_add(float * d_a, float * d_b, float * d_c, int size) {
  int global_id = (blockIdx.x * blockDim.x) + threadIdx.x;
  if (global_id % 4 == 0)
      d_a[global_id] = d_b[global_id] + d_c[global_id];
  else if (global_id % 4 == 1)
      d_a[global_id] = d_b[global_id] * d_c[global_id];
  else if (global_id % 4 == 2)
      d_a[global_id] = d_b[global_id] / d_c[global_id];
  else if (global_id % 4 == 3)
      d_a[global_id] = d_b[global_id] - d_c[global_id];
}
```

Stride Array:
First quarter: +
Second quarter: *
Third quarter: /
Fourth quarter: -

Consequences of Warps, Chunked Instructions

```
__global__ void vector_add_chunked(float * d_a, float * d_b, float * d_c, int size) {
  int global_id = (blockIdx.x * blockDim.x) + threadIdx.x;
  if (global_id < (size/4)*1)
      d_a[global_id] = d_b[global_id] + d_c[global_id];
  else if (global_id < (size/4)*2)
      d_a[global_id] = d_b[global_id] * d_c[global_id];
  else if (global_id < (size/4)*3)
      d_a[global_id] = d_b[global_id] / d_c[global_id];
  else if (global_id < (size/4)*4)
      d_a[global_id] = d_b[global_id] - d_c[global_id];
}
```

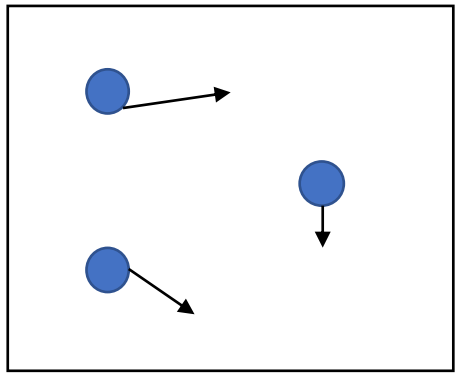
What is the right way to program GPUs?

- Still an open question!

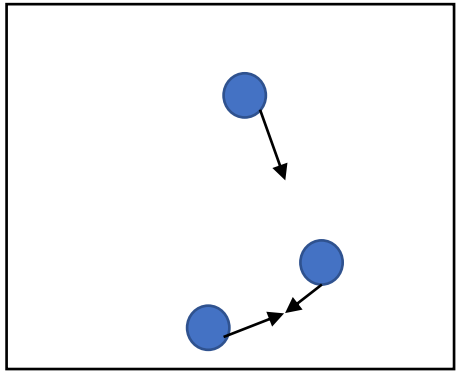
Homework 4 - first look

- Your assignment:
  - N-body simulation

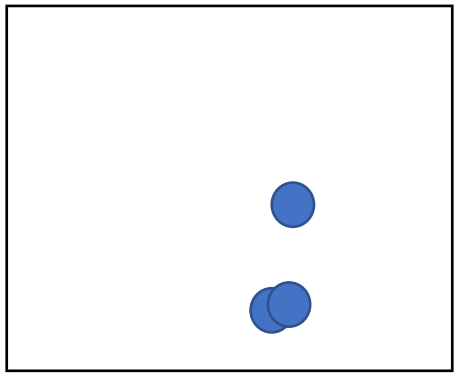
- Each particle interacts with every other particle



time = 0



time = 1

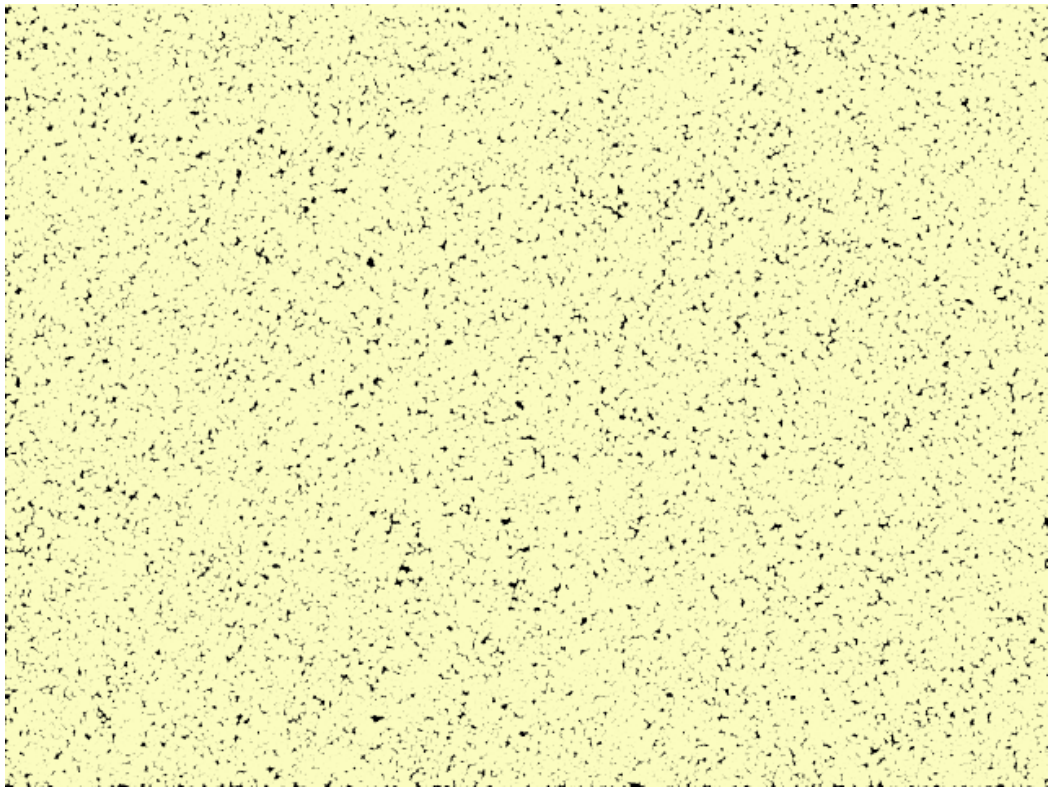


time = 2

Examples

- Gravity

- Boids (bird-oid object)
  An artificial life algorithm which simulates the flocking behavior of birds, and related group motion.
  https://en.wikipedia.org/wiki/Boids



Your homework

- N-body require a little bit of physics background so we will do something simpler.
  - If you want to explore with physics, please feel free

- **Local attraction clustering**:
  - For each particle: find your closest neighbor
  - You can take one step in the x direction and one step in the y direction towards your closest neighbor.